19 The Einstein Equations

19.1 Heuristics

We expect the gravitational field equations to be non-linear second order partial differential equations for the metric. If we knew more about the weak field equations of gravity (which should thus be valid near the origin of an inertial coordinate system) we could use the Einstein equivalence principle (or the principle of general covariance) to deduce the equations for strong fields.

However, we do not know a lot about gravity beyond the Newtonian limit of weak time-independent fields and low velocities, simply because gravity is so 'weak'. Hence, we cannot find the gravitational field equations in a completely systematic way and some guesswork will be required.

Nevertheless we will see that with some very natural assumptions (and the benefit of hindsight) we will arrive at an essentially unique set of equations. Further theoretical (and aesthetical) confirmation for these equations will then come from the fact that they turn out to be the Euler-Lagrange equations of the absolutely simplest action principle for the metric imaginable.

Recall that, way back, in section 1.1, we had briefly discussed the possibility of a scalar relativistic theory of gravity described by an equation of the form (1.3)

$$\Delta\phi = 4\pi G_N \rho \quad \longrightarrow \quad \Box\phi = 4\pi G_N \rho \ . \tag{19.1}$$

We had noted there that one way to render this equation (tensorially) consistent is to think of both the left and the right hand side as (00)-components of some tensor, which we expressed in (1.6) as

$$\{\text{Some tensor generalising } \Delta\phi\}_{\alpha\beta} \sim 4\pi G_N T_{\alpha\beta} \ . \tag{19.2}$$

While this appeared to be an exotic proposal back in section 1.1, we now understand that this is exactly what is required, and we have a fairly precise idea of what this tensor on the left-hand side should be.

Indeed, recall from our discussion of the Newtonian limit of the geodesic equation that the weak static field produced by a non-relativistic mass density ρ is

$$g_{00} = -(1 + 2\phi) \ , \tag{19.3}$$

With the identification

$$T_{00} = \rho \ , \tag{19.4}$$

the Newtonian field equation $\Delta\phi = 4\pi G_N \rho$ can now also be written as

$$\Delta g_{00} = -8\pi G_N T_{00} \ . \tag{19.5}$$

This suggests that the weak-field equations for a general energy-momentum tensor take the form

$$E_{\alpha\beta} = \quad \{\text{Some tensor generalising } (-\Delta g_{00})\}_{\alpha\beta} \quad = 8\pi G_N T_{\alpha\beta} \ , \tag{19.6}$$

where $E_{\alpha\beta}$ is constructed from the metric and its first and second derivatives.

By the Einstein equivalence principle, if this equation is valid for weak fields (i.e. near the origin of an inertial coordinate system) then also the equations which govern gravitational fields of arbitrary strength must be of this form, with $E_{\mu\nu}$ a tensor constructed from the metric and its first and second derivatives.

Another way of anticipating what form the field equations for gravity may take is via an analogy, a comparison of the geodesic deviation equations in Newton's theory and in General Relativity. Recall that in Newton's theory we have

$$\begin{aligned} \frac{d^2}{dt^2}\delta x^i &= -K^i{}_j \delta x^j \\ K^i{}_j &= \partial^i \partial_j \phi \ , \end{aligned} \tag{19.7}$$

whereas in General Relativity we have

$$\begin{aligned} (D_\tau)^2 \delta x^\mu &= -K^\mu_{\ \nu} \delta x^\nu \\ K^\mu_{\ \nu} &= R^\mu_{\ \lambda\nu\rho} \dot{x}^\lambda \dot{x}^\rho \ . \end{aligned} \tag{19.8}$$

Now Newton's field equation is

$$\mathrm{Tr}\, K \equiv \Delta\phi = 4\pi G_N \rho \ , \tag{19.9}$$

while in General Relativity we have

$$\mathrm{Tr}\, K = R_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \ . \tag{19.10}$$

This suggests that somehow in the gravitational field equations of General Relativity, $\Delta\phi$ should be replaced by the Ricci tensor $R_{\mu\nu}$. Note that, at least roughly, the tensorial structure of this identification is compatible with the relation between ϕ and g_{00} in the Newtonian limit, the relation between ρ and the 0-0 component T_{00} of the energy momentum tensor, and the fact that for small velocities $R_{\mu\nu}\dot{x}^\mu\dot{x}^\nu \sim R_{00}$.

We will now turn to a somewhat more precise argument along these lines which will enable us to determine $E_{\mu\nu}$.

19.2 More Systematic Approach

Let us take stock of what we know about $E_{\mu\nu}$.

1. $E_{\mu\nu}$ is a tensor

2. $E_{\mu\nu}$ has the dimensions of a second derivative. If we assume that no new dimensionful constants enter in $E_{\mu\nu}$ then it has to be a linear combination of terms which are either linear in second derivatives of the metric or quadratic in the first derivatives of the metric. (Later on, we will see that there is the possibility of a zero derivative term, but this requires a new dimensionful constant, the *cosmological constant* Λ. Higher derivative terms or higher non-linearities could in principle appear but would only be relevant at very high energies.)

3. $E_{\mu\nu}$ is symmetric since $T_{\mu\nu}$ is symmetric.

4. Since $T_{\mu\nu}$ is covariantly conserved, the same has to be true for $E_{\mu\nu}$,

$$\nabla_\mu T^{\mu\nu} = 0 \Rightarrow \nabla_\mu E^{\mu\nu} = 0 \ . \tag{19.11}$$

5. Finally, for a weak static gravitational field and non-relativistic matter we should find

$$E_{00} = -\Delta g_{00} \ . \tag{19.12}$$

Now it turns out that these conditions (1)-(5) determine $E_{\mu\nu}$ *uniquely*! First of all, (1) and (2) tell us that $E_{\mu\nu}$ has to be a linear combination

$$E_{\mu\nu} = aR_{\mu\nu} + bg_{\mu\nu}R \ , \tag{19.13}$$

where $R_{\mu\nu}$ is the Ricci tensor and R the Ricci scalar. Then condition (3) is automatically satisfied.

To implement (4), we rewrite the above as a linear combination of the Einstein tensor (8.97) and $g_{\mu\nu}R$,

$$E_{\mu\nu} = aG_{\mu\nu} + cg_{\mu\nu}R \equiv a(R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R) + cg_{\mu\nu}R \tag{19.14}$$

and recall the contracted Bianchi identity (8.95,8.96),

$$\nabla^\mu G_{\mu\nu} = 0 \ . \tag{19.15}$$

It follows that (4) is satisfied iff $c\nabla_\nu R = c\partial_\nu R = 0$. We therefore have to require either $\nabla_\nu R = 0$ or $c = 0$. That the first possibility is ruled out (inconsistent) can be seen by taking the trace of (19.6),

$$E^\mu_{\ \mu} = (4c - a)R = 8\pi G_N T^\mu_{\ \mu} \ . \tag{19.16}$$

Thus, R is proportional to $T^\mu_{\ \mu}$ and since this quantity need certainly not be constant for a general matter configuration, we are led to the conclusion that $c = 0$. Thus we find

$$E_{\mu\nu} = aG_{\mu\nu} \ . \tag{19.17}$$

We can now use condition (5) to determine the constant a.

By the above considerations we have determined the field equations to be of the form

$$aG_{\mu\nu} = 8\pi G_N T_{\mu\nu} \quad , \tag{19.18}$$

with a some, as yet undetermined, constant. We will now consider the Newtonian weak-field limit of this equation. We need to find that G_{00} is proportional to Δg_{00} and we can then use the condition (5) to fix the value of a. The following manipulations are somewhat analogous to those we performed in section 3.3 when considering the Newtonian limit of the geodesic equation. The main difference is that now we are dealing with second derivatives of the metric rather than with just its first derivatives entering in the geodesic equation.

As in section 3.3, let us begin by stating the assumptions that we make when considering the Newtonian limit:

1. Weak Fields

   We take this to mean that there exists a coordinate system $x^\alpha = (x^0, x^i)$ in which the metric takes the form

   $$g_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta} \tag{19.19}$$

   with $\eta_{\alpha\beta}$ the standard form of the Minkowski metric and $h_{\alpha\beta}$ and its derivatives small. In practice this means that in the following we will neglect terms that are quadratic or of higher order in $h_{\alpha\beta}$.

2. Time-independent Fields

   We assume that in these coordinates the gravitational field is time-independent, i.e. that one has

   $$\partial_0 g_{\alpha\beta} = \partial_0 h_{\alpha\beta} = 0 \quad . \tag{19.20}$$

3. Non-relativistic Matter Source

   This replaces the condition that particles move non-relativistically (with coordinate speeds $v \ll c$), and we take this to mean that the only non-negligible contribution to the energy-momentum tensor $T_{\alpha\beta}$ comes from the energy density $T_{00} = \rho$,

   $$T_{00} = \rho \neq 0 \quad , \quad T_{\alpha\beta} = 0 \quad \text{otherwise} \quad . \tag{19.21}$$

So we need to determine

$$G_{00} = R_{00} - \tfrac{1}{2} g_{00} R \quad . \tag{19.22}$$

Since the scalar curvature is at least linear in $h_{\alpha\beta}$, to leading order in $h_{\alpha\beta}$ we can replace $g_{00} \to \eta_{00} = -1$ to obtain

$$G_{00} = R_{00} + \tfrac{1}{2} R \quad , \tag{19.23}$$

where here and in the remainder of this calculation equality signs signify equalities to leading order in $h_{\alpha\beta}$.

To bootstrap the calculation of G_{00}, we start from

$$T_{ij} = 0 \quad \Rightarrow \quad G_{ij} = 0 \quad \Leftrightarrow \quad R_{ij} = \tfrac{1}{2} g_{ij} R \ , \tag{19.24}$$

and by the same reasoning as above we can write this as

$$R_{ij} = \tfrac{1}{2} \delta_{ij} R \ . \tag{19.25}$$

Therefore, for the scalar curvature we find

$$R = g^{\alpha\beta} R_{\alpha\beta} = \eta^{\alpha\beta} R_{\alpha\beta} = -R_{00} + \delta^{ij} R_{ij} = -R_{00} + \tfrac{3}{2} R \ , \tag{19.26}$$

or

$$R = 2R_{00} \ . \tag{19.27}$$

Thus

$$G_{00} = R_{00} + \tfrac{1}{2} R = 2R_{00} \tag{19.28}$$

and it just remains to calculate this one component of the Ricci tensor. In the weak field limit, R_{00} is given by

$$R_{00} = R^{k}_{\ 0k0} = \delta^{ik} R_{i0k0} \ . \tag{19.29}$$

Moreover, in this limit only the linear (second derivative) part of $R_{\mu\nu\lambda\sigma}$ will contribute, not the terms quadratic in first derivatives. Thus we can use the expression (8.13) for the curvature tensor. Additionally, in the static case we can ignore all time derivatives. Then only one term (the third) of (8.13) contributes and we find

$$R_{i0k0} = -\tfrac{1}{2} g_{00,ik} \ , \tag{19.30}$$

and therefore

$$R_{00} = -\tfrac{1}{2} \Delta g_{00} \ . \tag{19.31}$$

Putting everything together, we get

$$G_{00} = 2R_{00} = -\Delta g_{00} \ . \tag{19.32}$$

Thus we obtain the correct functional form of E_{00} and comparison with condition (5) determines $a = +1$ and therefore $E_{\alpha\beta} = G_{\alpha\beta}$. See also section 23.3 for a somewhat more streamlined and covariant derivation of this result from the linearised Einstein equations.

19.4 Einstein Equations

We have finally arrived at the Einstein equations for the gravitational field (metric) of a matter-energy configuration described by the energy-momentum tensor $T_{\mu\nu}$. It is

$$\boxed{R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R = 8\pi G_N \, T_{\mu\nu}} \tag{19.33}$$

These are the equations that replace the Newtonian (Poisson) equation for the gravitational potential.

Another common way of writing the Einstein equations is obtained by taking the trace of (19.33), which yields

$$R - 2R = 8\pi G_N \, T^{\mu}_{\mu} \equiv 8\pi G_N \, T \quad , \tag{19.34}$$

and substituting this back into (19.33) to obtain

$$R_{\mu\nu} = 8\pi G_N (T_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} T) \quad . \tag{19.35}$$

In particular, for the vacuum, $T_{\mu\nu} = 0$, the Einstein equations are

$$T_{\mu\nu} = 0 \quad \Rightarrow \quad R_{\mu\nu} = 0 \quad , \tag{19.36}$$

and this condition is equivalent to the vanishing of the Einstein tensor,

$$G_{\mu\nu} = 0 \quad \Leftrightarrow \quad R_{\mu\nu} = 0 \quad . \tag{19.37}$$

A space-time metric satisfying this equation is, for obvious reasons, said to be Ricci flat.

A priori, the Einstein equations constitute 10 coupled non-linear (actually quasi-linear, since they are linear in second derivatives) second order partial differential equations for the metric $g_{\mu\nu}(x)$, which appears both in the Einstein tensor on the left-hand side of these equations as well as usually also on the right-hand side in the matter energy-momentum tensor (we will see below, in section 19.7, that these 10 equations are linked by 4 differential identities, the contracted Bianchi identities).

This is a tremendously complicated set of equations, and trying to learn and say something about general properties of solutions to these equations is very challenging.[34]

[34] The current state of knowledge and understanding of the mathematical structure of the Einstein equations, in particular regarding the properties of the Cauchy (initial value) problem for the Einstein equations, is described in detail in the awe-inspiring œuvre *General Relativity and the Einstein Equations* by Y. Choquet-Bruhat (warning: not for the faint of heart). A readable historical introduction to the Cauchy problem for the Einstein equations is given by her in Y. Choquet-Bruhat, *Beginnings of the Cauchy problem*, `arXiv:1410.3490 [gr-qc]`.

Even the vacuum Einstein equations still constitute a complicated set of non-linear coupled partial differential equations whose general solution is not, and probably will never be, known. Usually one makes some assumptions, in particular regarding the symmetries of the metric, that reduce the number of independent variables from 10 functions $g_{\alpha\beta}(x)$ of 4 variables to a smaller number of functions depending on a smaller number of variables, and which then simplify the equations to the extent that they can be analysed explicitly, either analytically, or at least qualitatively or numerically. How to do this in practice (in the simplest non-trivial situations), will be explained in detail later on in these notes.

REMARKS:

1. With c not set equal to one, and with the convention that T_{00} is normalised such that it gives the energy-density rather than the mass-density, one finds that the factor $8\pi G_N$ on the right hand side should be replaced by

$$8\pi G_N \quad \to \quad \frac{8\pi G_N}{c^4} \ . \tag{19.38}$$

A note on dimensions: Newton's constant has dimensions (M mass, L length, T time) $[G_N] = \mathsf{M}^{-1}\mathsf{L}^3\mathsf{T}^{-2}$ so that

$$[G_N] = \mathsf{M}^{-1}\mathsf{L}^3\mathsf{T}^{-2} \quad \Rightarrow \quad [G_N/c^4] = \mathsf{L}^{-1}\mathsf{M}^{-1}\mathsf{T}^2 \ . \tag{19.39}$$

Moreover, an energy density $\rho = \mu c^2$, μ a mass density, has dimensions

$$[\rho] = [\mu c^2] = \mathsf{M}\mathsf{L}^{-3}\mathsf{L}^2\mathsf{T}^{-2} = \mathsf{M}\mathsf{L}^{-1}\mathsf{T}^{-2} \ . \tag{19.40}$$

Thus

$$[\rho G_N/c^4] = \mathsf{L}^{-2} = [R_{\mu\nu}] \ , \tag{19.41}$$

as it should be. Frequently, an alternative (and equally reasonable) convention is used in which T_{00} is a mass density, so that then $T_{tt} = c^2 T_{00}$ is the energy density. In that case, the factor on the right-hand side of the Einstein equations is $8\pi G_N/c^2$.

2. The streamlined "derivation" of the Einstein equations given here may give the misleading impression that also for Einstein this was a straighforward affair. Nothing could be further from the truth. Not only do we have the benefit of hindsight. We also have a much more systematic and advanced understanding of Riemannian geometry and tensor calculus than was available to Einstein at the time. This concerns in particular things like the contracted Bianchi identities and their importance for energy-momentum (non-)conservation and for general covariance (to be briefly discussed in section 19.7 below).[35]

[35] For an illuminating brief account of the torturous and convoluted route and crucial final stages that led Einstein (and Grossmann) to the correct field equations, see N. Straumann, *Einstein's 'Zürich Notebook' and his Journey to General Relativity*, `arXiv:1106.0900v1 [physics.hist-ph]`.

3. As an aside, note that the trace (19.34) of the Einstein equations

$$R = -8\pi G_N \, T \quad , \tag{19.42}$$

is a scalar generally covariant differential equation for the metric (but it is of course far from sufficient to determine 10 independent components of the metric up to coordinate transformations). If one assumes, however, that the space-time metric can be parametrised by a single scalar ψ, say (somewhat like in the Newtonian limit), e.g. by stipulating that it only differs from the Minkowski metric by a conformal factor, as in

$$g_{\mu\nu} = \psi^2 \eta_{\mu\nu} \quad , \tag{19.43}$$

then a scalar equation like (19.42) (the numerical constant needs to be adjusted appropriately in order to obtain the correct Newtonian limit) provides a differential equation for ψ and thus a generally covariant scalar theory of gravity. A theory of this kind, a geometrisation and covariantisation of previous scalar theories of gravity, was proposed by Einstein and Fokker in 1913/14, some two years before Einstein arrived at the final (tensorial) form of the field equations. In this theory, there is no coupling of gravity and Maxwell theory (which has a traceless energy-momentum tensor), and null lines are identical to null lines in Minkowski space (because of conformal flatness), so for either of these reasons there is no bending of lightrays by the gravitational field in such a theory.

4. As we saw before, in two and three dimensions, vanishing of the Ricci tensor implies the vanishing of the Riemann tensor. Thus in these cases, space-times are necessarily flat away from where there is matter, i.e. at points at which $T_{\mu\nu}(x) = 0$. Thus there are no true gravitational fields and no gravitational waves.

   In four dimensions, however, the situation is completely different. As we saw, the Ricci tensor has 10 independent components whereas the Riemann tensor has 20. Thus there are 10 components of the Riemann tensor which can curve the vacuum, as e.g. in the field around the sun, and a lot of interesting physics is already contained in the vacuum Einstein equations.

5. If for whatever reason one is interested in studying solutions to the matter + Einstein equations in dimensions other than $D = 3 + 1$, this is straightforward and there are just a a few small points to pay attention to:

   - The Einstein tensor, i.e. the (unique) rank-2 tensor that can be constructed from the Riemann curvature tensor which has vanishing covariant divergence, has the same form in any dimension, $G_{\mu\nu} = R_{\mu\nu} - (1/2)g_{\mu\nu}R$.
   - Likewise, what will appear on the right-hand side of the equations is the appropriate generally covariant energy-momentum tensor.

- However, in the higher-dimensional analogue of the Einstein equations the constant of proportionality between the Einstein and energy-momentum tensors should not be called $8\pi G_N$. After all, this factor was determined from the Newtonian limit of the $(3+1)$-dimensional Einstein equations where e.g. a factor 4π has, via the Poisson equation for a point mass, its origin in the fact that the area of a unit 2-sphere is 4π. Thus we will just call it κ (which is then related in a dimension-dependent way to however one wants to normalise the D-dimensional gravitational coupling constant). Thus precisely as in 4 dimensions one can write the Einstein equations as

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}\, g_{\mu\nu} R = \kappa\, T_{\mu\nu} \quad . \tag{19.44}$$

- If one wants to use the analogue of (19.35), one should pay attention to the fact that it is less symmetric with respect to (19.44) than its 4-dimensional counterpart since in $D = n + 1$ dimensions it takes the form

$$R_{\mu\nu} = \kappa(T_{\mu\nu} - \frac{1}{D-2}\, g_{\mu\nu} T^{\lambda}_{\ \lambda}) \quad . \tag{19.45}$$

19.5 Cosmological Constant

As mentioned before, there is one more term that can be added to the Einstein equations provided that one relaxes the condition (2) that only terms quadratic in derivatives should appear. This term takes the form $\Lambda g_{\mu\nu}$. This is compatible with the condition (4) (the conservation law) provided that Λ is a constant, the *cosmological constant*. It is a dimensionful parameter with dimension $[\Lambda] = \mathsf{L}^{-2}$ one over length squared.

The Einstein equations with a cosmological constant now read

$$R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = 8\pi G_N T_{\mu\nu} \quad . \tag{19.46}$$

To be compatible with condition (5) ((1), (3) and (4) are obviously satisfied), Λ has to be quite small (and observationally it is very small indeed).

Remarks:

1. The vacuum Einstein equations with a cosmological constant read

$$R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R = -\Lambda g_{\mu\nu} \quad . \tag{19.47}$$

   Taking traces, this implies (and is equivalent to)

$$R_{\mu\nu} = \Lambda g_{\mu\nu} \quad , \tag{19.48}$$

   which is the counterpart of the Ricci-flatness condition for vacuum solutions of the Einstein equations without a cosmological constant. In general, solutions to the equation $R_{\mu\nu} = c g_{\mu\nu}$ for some constant c (and either Riemannian or Lorentzian signature) are known as *Einstein manifolds* in the mathematics literature.

2. Λ gives a contribution to the energy-momentum tensor that, in Minkowski space, would be proportional to the Minkowski metric and Lorentz-invariant, thus compatible with the symmetries of the vacuum, and Λ is often said to play the role of a vacuum energy density (more precisely vacuum energy should perhaps be considered as one possible contribution to the cosmological constant - see section 38.4 for further discussion of this issue).

3. Comparing with the energy-momentum tensor of, say, a perfect fluid (see (7.70) in section 7.5 or section 35.2),

$$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + p g_{\mu\nu} \quad , \tag{19.49}$$

we see that Λ corresponds to the energy density and pressure values

$$\rho_\Lambda = -p_\Lambda = \frac{\Lambda}{8\pi G_N} \quad , \tag{19.50}$$

and to an energy-momentum tensor

$$T^\Lambda_{\mu\nu} = -\rho_\Lambda g_{\mu\nu} \quad . \tag{19.51}$$

Thus, depending on the sign of Λ either the energy density or the pressure is negative,

$$\Lambda < 0 \quad \Rightarrow \quad \rho_\Lambda < 0 \quad , \quad \Lambda > 0 \quad \Rightarrow \quad p_\Lambda < 0 \quad . \tag{19.52}$$

4. The cosmological constant was originally introduced by Einstein because he was unable to find static cosmological solutions without it. We will review this *Einstein Static Universe* in section 37.2. After Hubble's discovery of the expansion of the universe, a static universe fell out of fashion and the cosmological constant was no longer required.

5. However, things are not as simple as that. Just because it is not required does not mean that it is not there. In fact, one of the biggest puzzles in theoretical physics today is why the cosmological constant is so small. According to standard quantum field theory lore, the vacuum energy density should be many many orders of magnitude larger than astrophysical observations allow. Now usually in quantum field theory one does not worry too much about the vacuum energy as one can normal-order it away. However, as we know, gravity is unlike any other theory in that not only energy-differences but absolute energies matter (and cannot just be dropped).

   The question why the observed cosmological constant is so small (and recent astrophysical observations appear to favour a tiny non-zero value) is one aspect of what is known as the *Cosmological Constant Problem*. See section 38.4 for a brief discussion of this profound issue and some references.

6. We will consider the possibility that $\Lambda \neq 0$ only in the sections on cosmology (in all other applications, Λ can indeed be neglected).

The Einstein equations

$$G_{\mu\nu} = 8\pi G_N \, T_{\mu\nu} \tag{19.53}$$

can, taken at face value, be regarded as ten algebraic equations for certain traces of the Riemann tensor $R_{\mu\nu\rho\sigma}$. $R_{\mu\nu\rho\sigma}$ has, as we know, twenty independent components, so how are the other ten determined? The obvious answer, already given above, is of course that we solve the Einstein equations for the metric $g_{\mu\nu}$ and then calculate the Riemann curvature tensor of that metric.

However, this answer leaves something to be desired because it does not really provide an explanation of how the information about these other components is encoded in the Einstein equations. It is interesting to understand this because it is precisely these components of the Riemann tensor wich represent the effects of gravity in vacuum, i.e. where $T_{\mu\nu} = 0$, like tidal forces and gravitational waves.

The more insightful answer is that the information is encoded in the Bianchi identities which serve as propagation equations for the trace-free parts of the Riemann tensor away from the regions where $T_{\mu\nu} \neq 0$.

Let us see how this works. Recall from section 11.4 the decomposition of the Riemann tensor into the traceless Weyl tensor and the trace parts, the Ricci tensor and Ricci scalar,

$$\begin{aligned} C_{\mu\nu\rho\sigma} = {} & R_{\mu\nu\rho\sigma} \\ & - \frac{1}{D-2}(g_{\mu\rho}R_{\nu\sigma} + R_{\mu\rho}g_{\nu\sigma} - g_{\nu\rho}R_{\mu\sigma} - R_{\nu\rho}g_{\mu\sigma}) \\ & + \frac{1}{(D-1)(D-2)}R(g_{\mu\rho}g_{\nu\sigma} - g_{\nu\rho}g_{\mu\sigma}) \ . \end{aligned}$$

In the vacuum, $R_{\mu\nu} = 0$, and therefore at points where the energy-momentum tensor vanishes one has

$$T_{\mu\nu}(x) = 0 \quad \Rightarrow \quad R_{\mu\nu\rho\sigma}(x) = C_{\mu\nu\rho\sigma}(x) \ . \tag{19.54}$$

As anticipated, the Weyl tensor thus encodes the information about the gravitational field in vacuum.

The question thus is how $C_{\mu\nu\rho\sigma}$ is determined everywhere in space-time by an energy-momentum tensor which may be localised in some finite region of space-time. To address that question we make use of the relation (11.88) derived in section 11.4,

$$\nabla^\mu C_{\mu\nu\rho\sigma} = (D-3)\,(\nabla_\rho P_{\nu\sigma} - \nabla_\sigma P_{\nu\rho}) \ . \tag{19.55}$$

Here $P_{\mu\nu}$ is the Schouten tensor (11.81),

$$P_{\mu\nu} = \frac{1}{D-2}\left(R_{\mu\nu} - \frac{1}{2(D-1)}g_{\mu\nu}R\right) \ . \tag{19.56}$$

Using the D-dimensional Einstein equations (19.44), (19.45) to replace the Ricci tensor and Ricci scalar by the energy-momentum tensor, one now obtains a propagation equation for the Weyl tensor of the form

$$\nabla^\mu C_{\mu\nu\rho\sigma} = J_{\nu\rho\sigma} \quad , \tag{19.57}$$

where $J_{\nu\rho\sigma}$ depends only on the energy-momentum tensor and its derivatives,

$$J_{\nu\rho\sigma} = \kappa \, \frac{D-3}{D-2} \left[\nabla_\rho T_{\nu\sigma} - \nabla_\sigma T_{\nu\rho} - \frac{1}{D-1} \left[\nabla_\rho T^\lambda{}_\lambda g_{\nu\sigma} - \nabla_\sigma T^\lambda{}_\lambda g_{\nu\rho} \right] \right] \quad . \tag{19.58}$$

This is the equation which determines the Weyl tensor components in terms of the sources. It is reminiscent of the Maxwell equation

$$\nabla^\mu F_{\mu\nu} = -J_\nu \quad , \tag{19.59}$$

and provides an intuitive (as well as, if required, detailed analytical) understanding of the propagation properties of the gravitational field.

19.7 General Covariance and Significance of the Bianchi Identities

Let us try to understand in a bit more detail, but necessarily at a very superficial and unsophisticated level, the structure of the Einstein equations.

As a first step, let us do something that we should have perhaps done rightaway, namely count the number of dynamical variables and the number of equations we have:

- the dynamical variables are the components $g_{\alpha\beta}(x)$ of the metric, i.e. 10 functions of 4 variables.
- the Ricci or Einstein tensor is symmetric; therefore the Einstein equations consitute a set of ten algebraically independent second order differential equations for the metric $g_{\alpha\beta}$.

At first, this "ten 2nd order equations for ten unknowns" looks exactly right: specifying the values of the metric and its first time-derivative as initial values on some (constant "time") hypersurface, say, this should then uniquely determine the ten components of the metric in some region to the future of that hypersurface.

At second sight, however, this cannot possibly be right and the end of the story and, if true, would actually be a major disaster. After all, the Einstein equations are generally covariant. Thus, given one metric that is a solution to the Einstein equations, one should be able to perform an arbitrary coordinate transformation and still have a (physically equivalent) solution to the Einstein equations. That means that the (ten?) Einstein equations should not determine the ten components of the metric uniquely but only

up to arbitrary coordinate transformations, i.e. up to four arbitrary functions of four variables.

Phrased in terms of initial values, one should be able to perform arbitrary time-dependent coordinate transformations on a solution, but if these coordinate transformations happen to be the identity transformation on the initial hypersurface, then these solutions related by (future) coordinate transformations should arise from the same initial data.

Either way we should expect only six independent generally covariant equations for the metric, determining the 10 components of the metric up to 4 arbitary functions. How does that happen? Here we should recall the contracted Bianchi identities. They tell us that

$$\nabla^\alpha G_{\alpha\beta} = 0 \ . \tag{19.60}$$

We see that, even though the ten Einstein equations are algebraically independent, there are actually four differential relations among them, so this is just right.

It is no coincidence, by the way, that the Bianchi identities come to the rescue of general covariance. We will see in section 20.6 that the Bianchi identities can in fact be understood as a consequence of the general covariance of the Einstein equations (and of the corresponding action principle).

The general covariance of the Einstein equations is reflected in the fact that only six of the ten equations are truly dynamical 2nd-order differential equations while four of them constrain the initial values of the fields on some spacelike hypersurface. Indeed, in terms of some choice $(x^\alpha) = (t, x^k)$ of time and space coordinates, the Bianchi identities

$$\nabla_\alpha G^{\alpha\beta} = \partial_\alpha G^{\alpha\beta} + \Gamma^\alpha_{\ \alpha\gamma} G^{\beta\gamma} + \Gamma^\beta_{\ \alpha\gamma} G^{\alpha\gamma} = 0 \tag{19.61}$$

can be written as

$$\partial_t G^{t\beta} = -\partial_k G^{k\beta} - \Gamma^\alpha_{\ \alpha\gamma} G^{\beta\gamma} - \Gamma^\beta_{\ \alpha\gamma} G^{\alpha\gamma} \ . \tag{19.62}$$

Since the 3 terms on the right-hand side contain at most 2nd time derivatives of the metric, the 4 components $G_{t\beta}$ of the Einstein tensor can contain at most 1st time derivatives of the metric. Thinking of initial data as being given by the metric and its 1st time-derivative on some initial hypersurface, this means that the components $G_{t\beta} = 0$ of the Einstein equations (or their counterpart in the presence of matter) impose constraints on these initial data and do not provide evolution equations for these initial data.

The perhaps more familiar counterpart of these constraints in the case of Maxwell theory is the Gauss Law constraint $\vec{\nabla}.\vec{E} = 0$, which arises as the 0-component of the Maxwell equations $\partial_a F^{ab} = 0$,

$$\partial_a F^{a0} = \partial_k F^{k0} = -\vec{\nabla}.\vec{E} = 0 \ , \tag{19.63}$$

and which also involves at most 1st time-derivatives of the dynamical field (the gauge field), and thus constitutes a constraint on the initial conditions rather than a true evolution equation.

In this case, the obvious ("contracted Bianchi") identity

$$\partial_a \partial_b F^{ab} = 0 \quad \Leftrightarrow \quad \partial_0(\partial_i F^{i0}) = -\partial_k(\partial_a F^{ak}) \tag{19.64}$$

implies

1. that the 4 Maxwell equations are not independent (as required by gauge invariance as they should only determine the 4 components A_a of the gauge field up to gauge transformations) and

2. that the Gauss Law contraint equation is "propagated", i.e. that by virtue of the true equations of motion it will hold at all times if it holds initially:

$$(\partial_a F^{ak})|_{t=0} = 0 \quad \Rightarrow \quad (\partial_k(\partial_a F^{ak}))_{t=0} = 0 \quad \Rightarrow \quad (\partial_0(\partial_i F^{i0}))_{t=0} = 0 \ , \tag{19.65}$$

   and likewise for the higer t-derivatives,

$$(\partial_0 \partial_a F^{ak})|_{t=0} = 0 \quad \Rightarrow \quad (\partial_k \partial_0(\partial_a F^{ak}))_{t=0} = 0 \quad \Rightarrow \quad (\partial_0^2(\partial_i F^{i0}))_{t=0} = 0 \ , \tag{19.66}$$

   etc. Thus if the true dynamical equations are satisfied at all times, the constraints will be satisfied at all times provided that they are satisfied initially.

Analogously, for the Einstein equations the contracted Bianchi identity in the form (19.62) implies not only 4 relations among the 10 field equations (as required by general covariance) but also that the constraints of general relativity are again "propagated" in this sense. One simple way to see this (or that this is plausible, at least - in order to prove a theorem one would need te be more precise about the initial value formulation and make sure that it leads to a well-defined time evolution etc.) is to note that by (19.62) $G^{\alpha\beta} = 0$ at $t = t_0$ (thus also $\partial_k G^{\alpha\beta} = 0$ at $t = t_0$) implies

$$G^{\alpha\beta}|_{t=t_0} = 0 \quad \Rightarrow \quad (\partial_t G^{t\beta})|_{t=t_0} = 0 \tag{19.67}$$

(and likewise for higher t-derivatives).

We will discuss this and related issues in some more detail from a slightly different (Hamiltonian) perspective in section 21.

20 Einstein Equations from an Action Principle

20.1 Einstein-Hilbert Action

To increase our confidence that the Einstein equations we have derived above are in fact reasonable and almost certainly correct, we can adopt a more modern point of view. We can ask if the Einstein equations follow from an action principle or, alternatively, what would be a natural action principle for the metric.

After all, for example in the construction of the Standard Model, one also does not start with the equations of motion but one writes down the simplest possible Lagrangian with the desired field content and symmetries.

We will start with the gravitational part, i.e. the Einstein tensor $G_{\alpha\beta}$ of the Einstein equations, and deal with the matter part, the energy-momentum tensor $T_{\alpha\beta}$, later.

By general covariance, an action for the metric $g_{\alpha\beta}$ will have to take the form

$$S = \int \sqrt{g} d^4x \; \Phi(g_{\alpha\beta}) \;\; , \tag{20.1}$$

where Φ is a scalar constructed from the metric. So what is Φ going to be? Clearly, the simplest choice is the Ricci scalar R, and this is also the unique choice if one is looking for a scalar constructed from not higher than second derivatives of the metric. Therefore we postulate the beautifully simple and elegant action

$$\boxed{S_{EH}[g_{\alpha\beta}] = \int \sqrt{g} d^4x \; R} \tag{20.2}$$

known as the Einstein-Hilbert action. It was presented by Hilbert practically on the same day that Einstein presented his final form (19.33) of the gravitational field equations. Discussions regarding who did what first and who deserves credit for what have been a favourite occupation of historians of science ever since. However, Hilbert's work would certainly not have been possible without Einstein's realisation that gravity should be regarded not as a force but as a property of space-time and his physical insight that Riemannian geometry and tensor analysis provide the correct framework for embodying the equivalence principle. Regarding the action principle for general relativity, in his superb scientific biography of Einstein, A. Pais says

> Hilbert was not the first to apply this principle to gravitation. Lorentz had done it before him. So had Einstein, a few weeks earlier. Hilbert was the first, however, to state this principle correctly.[36]

[36] A. Pais, *Subtle is the Lord* (chapter 14.d, which also contains a detailed account of the interaction between Einstein and Hilbert in the crucial November 1915 period).

We will now prove that the Euler-Lagrange equations following from the Einstein-Hilbert Lagrangian indeed give rise to the Einstein tensor and the vacuum Einstein equations. It is truly remarkable, that such a simple Lagrangian is capable of explaining practically all known gravitational, astrophysical and cosmological phenomena.

Before turning to a proof of this statement, I need to make one preliminary remark:

In this discussion we will at first ignore total derivative (or boundary) terms that one picks up from integration by parts of the variations and concentrate on the bulk Euler-Lagrange equations of motion. In standard variational problems one usually justifies this by appealing to the fact that one can e.g. choose the variations of the fields to vanish on the boundary and that therefore such boundary terms are zero. In the case at hand, things are a bit more complicated since the boundary terms that one picks up in the process of performing the variations turns out to depend both on the variation of the field (i.e. the metric) on the boundary and on its normal derivative, and it is not consistent to require both to be zero (i.e. to impose both Dirichlet and Neumann boundary conditions). This whole issue is interesting in its own right and warrants a separate discussion, and therefore we will deal with it afterwards, in sections 20.4 and 20.5.

Returning to the Einstein-Hilbert action, we now need to determine its behaviour under a variation of the metric. Since the Ricci scalar is $R = g^{\alpha\beta}R_{\alpha\beta}$, it turns out to be more convenient to consider variations $\delta g^{\alpha\beta}$ of the inverse metric instead of $\delta g_{\alpha\beta}$. This is of course equivalent, the two variations being related by

$$\delta(g^{\alpha\beta}g_{\beta\gamma}) = \delta(\delta^{\alpha}_{\ \gamma}) = 0 \quad \Rightarrow \quad \delta g^{\alpha\beta} = -g^{\alpha\gamma}(\delta g_{\gamma\delta})g^{\delta\beta} \ . \tag{20.3}$$

Thus, as a first step we write

$$\begin{aligned} \delta S_{EH} &= \delta \int \sqrt{g} d^4x \; g^{\alpha\beta}R_{\alpha\beta} \\ &= \int d^4x \; \Big((\delta\sqrt{g}) g^{\alpha\beta}R_{\alpha\beta} + \sqrt{g}(\delta g^{\alpha\beta})R_{\alpha\beta} + \sqrt{g}g^{\alpha\beta}\delta R_{\alpha\beta} \Big) \ . \end{aligned} \tag{20.4}$$

Now we make use of the identity (5.77)

$$\delta\sqrt{g} = \tfrac{1}{2}\sqrt{g}g^{\alpha\beta}\delta g_{\alpha\beta} = -\tfrac{1}{2}\sqrt{g}g_{\alpha\beta}\delta g^{\alpha\beta} \ . \tag{20.5}$$

to deduce

$$\begin{aligned} \delta S_{EH} &= \int \sqrt{g} d^4x \; [(-\tfrac{1}{2}g_{\alpha\beta}R + R_{\alpha\beta})\delta g^{\alpha\beta} + g^{\alpha\beta}\delta R_{\alpha\beta}] \\ &= \int \sqrt{g} d^4x \; (R_{\alpha\beta} - \tfrac{1}{2}g_{\alpha\beta}R)\delta g^{\alpha\beta} + \int \sqrt{g} d^4x \; g^{\alpha\beta}\delta R_{\alpha\beta} \ . \end{aligned} \tag{20.6}$$

The first term all by itself would already give the Einstein tensor. Thus we need to show that the second term is a total derivative. I do not know of any particularly elegant

argument to establish this (in a coordinate basis - written in terms of differential forms this would be completely obvious), so this will require a little bit of work, but it is not difficult.

Postponing the proof of this statement to the next section 20.2, we have established that (ignoring boundary terms) the variation of the Einstein-Hilbert action gives the gravitational part (left hand side) of the Einstein equations,

$$\delta S_{EH}[g_{\alpha\beta}] = \delta \int \sqrt{g} d^4x \; R = \int \sqrt{g} d^4x \; G_{\alpha\beta} \delta g^{\alpha\beta} + \oint \dots \quad . \tag{20.7}$$

Remarks:

1. If one wants to include the cosmological constant Λ, then the action gets modified to

$$S_{EH,\Lambda} = \int \sqrt{g} d^4x \; (R - 2\Lambda) \quad . \tag{20.8}$$

Indeed, the only effect of including Λ is to replace $R \to R - 2\Lambda$ in the Einstein equations, so that

$$G_{\alpha\beta} = R_{\alpha\beta} - \tfrac{1}{2} g_{\alpha\beta} R \to R_{\alpha\beta} - \tfrac{1}{2} g_{\alpha\beta}(R - 2\Lambda) = G_{\alpha\beta} + \Lambda g_{\alpha\beta} \quad , \tag{20.9}$$

which gives rise to the modified Einstein equation (19.46).

2. Of course, once one is working at the level of the action, it is easy to come up with covariant generalisations of the Einstein-Hilbert action, such as

$$S = \int \sqrt{g} d^4x \; (R + c_1 R^2 + c_2 R_{\alpha\beta} R^{\alpha\beta} + c_3 R_{\alpha\beta\gamma\delta} R^{\alpha\beta\gamma\delta} + c_4 R \Box R + \dots) \quad , \tag{20.10}$$

with dimensionful coefficients c_k, but these invariably involve higher-derivative terms and/or higher non-linearities and are therefore irrelevant for low-energy physics and thus the world we live in. Such terms could be relevant for the early universe, however, and are also typically predicted by quantum theories of gravity like string theory.

3. A particular class of such higher-order actions has attracted some attention. As already briefly mentioned at the end of section 8.8, in $D > 4$ space-time dimensions there are other candidate tensors that could replace the Einstein tensor, provided that one is willing to give up linearity of the 2nd derivative terms of the metric. These tensors can be obtained from a variational principle involving very special linear combinations of higher order terms in the action, e.g. the *Gauss-Bonnet* term

$$L_{GB} = R^{\alpha\beta\gamma\delta} R_{\alpha\beta\gamma\delta} - 4 R^{\alpha\beta} R_{\alpha\beta} + R^2 \quad . \tag{20.11}$$

In $D = 4$ this term is (locally) a total derivative, and thus does not contribute to the equations of motion. It is non-trivial in $D > 4$, however, but nevertheless (a

priori totally non-obviously) leads to equations of motion that are no higher than 2nd order in derivatives.[37]

20.2 Appendix: A Formula for the Variation of the Ricci Tensor

The purpose of this technical appendix to the previous section is to derive a formula for the metric variation of the Ricci tensor which shows that indeed $g^{\alpha\beta}\delta R_{\alpha\beta}$ is a total derivative.

First of all, we need the explicit expression for the Ricci tensor in terms of the Christoffel symbols, which can be obtained by contraction of (8.5),

$$R_{\mu\nu} = \partial_\lambda \Gamma^\lambda_{\ \mu\nu} - \partial_\nu \Gamma^\lambda_{\ \mu\lambda} + \Gamma^\lambda_{\ \lambda\rho}\Gamma^\rho_{\ \nu\mu} - \Gamma^\lambda_{\ \nu\rho}\Gamma^\rho_{\ \lambda\mu} \ . \tag{20.12}$$

Now we need to calculate the variation of $R_{\mu\nu}$. We will not require the explicit expression in terms of the variations of the metric, but only in terms of the variations $\delta\Gamma^\mu_{\ \nu\lambda}$ induced by the variations of the metric. This simplifies things considerably.

Obviously, $\delta R_{\mu\nu}$ will then be a sum of six terms,

$$\delta R_{\mu\nu} = \partial_\lambda \delta\Gamma^\lambda_{\ \mu\nu} - \partial_\nu \delta\Gamma^\lambda_{\ \mu\lambda} + \delta\Gamma^\lambda_{\ \lambda\rho}\Gamma^\rho_{\ \nu\mu} + \Gamma^\lambda_{\ \lambda\rho}\delta\Gamma^\rho_{\ \nu\mu} - \delta\Gamma^\lambda_{\ \nu\rho}\Gamma^\rho_{\ \lambda\mu} - \Gamma^\lambda_{\ \nu\rho}\delta\Gamma^\rho_{\ \lambda\mu} \ . \tag{20.13}$$

Now the crucial observation is that $\delta\Gamma^\mu_{\ \nu\lambda}$ is a tensor. This follows from the arguments given in section 5.4, but I will repeat it here in the present context. Of course, we know that the Christoffel symbols themselves are not tensors, because of the inhomogeneous (second derivative) term appearing in the transformation rule under coordinate transformations, but this term is independent of the metric. Thus the metric variation of the Christoffel symbols indeed transforms as a tensor.

This can also be confirmed by explicit calculation. Just for the record, I will give an expression for $\delta\Gamma^\mu_{\ \nu\lambda}$ which is easy to remember as it takes exactly the same form as the definition of the Christoffel symbol, only with the metric replaced by the metric variation and the partial derivatives by covariant derivatives, i.e.

$$\delta\Gamma^\mu_{\ \nu\lambda} = \tfrac{1}{2} g^{\mu\rho}(\nabla_\nu \delta g_{\rho\lambda} + \nabla_\lambda \delta g_{\rho\nu} - \nabla_\rho \delta g_{\nu\lambda}) \ . \tag{20.14}$$

It turns out, none too surprisingly, that $\delta R_{\mu\nu}$ can be written rather compactly in terms of covariant derivatives of $\delta\Gamma^\mu_{\ \nu\lambda}$, namely as

$$\delta R_{\mu\nu} = \nabla_\lambda \delta\Gamma^\lambda_{\ \mu\nu} - \nabla_\nu \delta\Gamma^\lambda_{\ \mu\lambda} \ . \tag{20.15}$$

Thus one simply needs to replace the partial derivatives in (20.13) by covariant derivatives and drop the other terms that involve the undifferentiated Christoffel symbols.

[37] For a review of these so-called *Lanczos-Lovelock* models, see e.g. T. Padmanabhan, D. Kothawala, *Lanczos-Lovelock models of gravity*, `arXiv:1302.2151 [gr-qc]`.

In fact, this could not have been otherwise as (20.13) depends on the partial derivatives of the $\delta\Gamma$ but must at the same time be tensorial. The expression (20.15) is the unique possibility that fulfills these requirements. If you don't trust this argument (which essentially amounts to working at the origin of an inertial coordinate system where partial = covariant derivatives), you can also check this in detail (and thus perhaps in this way learn to trust and appreciate the quick argument):

As a first check on (20.15), note that the first term on the right hand side is manifestly symmetric and that the second term is also symmetric because of (5.49) and (6.71). To establish (20.15), one simply has to use the definition of the covariant derivative. The first term is

$$\nabla_\lambda \delta\Gamma^\lambda{}_{\mu\nu} = \partial_\lambda \delta\Gamma^\lambda{}_{\mu\nu} + \Gamma^\lambda{}_{\lambda\rho}\delta\Gamma^\rho{}_{\mu\nu} - \Gamma^\rho{}_{\mu\lambda}\delta\Gamma^\lambda{}_{\rho\nu} - \Gamma^\rho{}_{\nu\lambda}\delta\Gamma^\lambda{}_{\rho\mu} \quad , \tag{20.16}$$

which takes care of the first, fourth, fifth and sixth terms of (20.13). The remaining terms are

$$-\partial_\nu \delta\Gamma^\lambda{}_{\mu\lambda} + \delta\Gamma^\lambda{}_{\lambda\rho}\Gamma^\rho{}_{\nu\mu} = -\nabla_\nu \delta\Gamma^\lambda{}_{\mu\lambda} \quad , \tag{20.17}$$

which establishes (20.15).

What we really need is the contraction $g^{\mu\nu}\delta R_{\mu\nu}$, which we can now write as

$$\begin{aligned} g^{\mu\nu}\delta R_{\mu\nu} &= \nabla_\lambda(g^{\mu\nu}\delta\Gamma^\lambda{}_{\mu\nu}) - \nabla_\nu(g^{\mu\nu}\delta\Gamma^\lambda{}_{\lambda\mu}) \\ &= \nabla_\lambda\left(g^{\mu\nu}\delta\Gamma^\lambda{}_{\mu\nu} - g^{\mu\lambda}\delta\Gamma^\nu{}_{\nu\mu}\right) \quad . \end{aligned} \tag{20.18}$$

This establishes the claim that this term is a total derivative and hence gives rise to a boundary term in the variation of the Einstein-Hilbert action, a boundary term that does, however, require further discussion - see sections 20.4 and 20.5 below.

Using the explicit expression for $\delta\Gamma^\mu{}_{\nu\lambda}$ given above, we see that we can also write (20.18) rather neatly and explicitly as

$$\begin{aligned} g^{\mu\nu}\delta R_{\mu\nu} &= (\nabla^\mu\nabla^\nu - g^{\mu\nu}\Box)\delta g_{\mu\nu} \\ &= (g^{\mu\alpha}g^{\nu\beta} - g^{\mu\nu}g^{\alpha\beta})\nabla_\mu\nabla_\nu\delta g_{\alpha\beta} \\ &= \nabla_\lambda\left((g^{\lambda\alpha}g^{\nu\beta} - g^{\lambda\nu}g^{\alpha\beta})\nabla_\nu\delta g_{\alpha\beta}\right) \quad . \end{aligned} \tag{20.19}$$

This result will turn out to be useful on a couple of occasions later on in these notes, e.g. for the discussion of Noether currents associated to general covariance in section 20.6 and for the derivation of the energy-momentum tensor of a non-minimally coupled scalar field in section 22.3.

One can also use the identity (20.19) to rather painlessly determine the metric variation of some more complicated Lagrangians of the type (20.10). Consider e.g. the class of Lagrangians known as "$F(R)$ Lagrangians" where the Lagrangian is (none too surprisingly) some function $F(R)$ of the scalar curvature R,

$$S = \int \sqrt{g}d^4x\ F(R) \tag{20.20}$$

(for no particularly compelling reason, at least as far as I can see ("it can be done" is *not* a compelling reason ...), a lot of work has been dedicated to such Lagrangians in the last ten years, as a quick look at the `arXiv` will reveal).

The metric variation of this action is evidently

$$\delta S = \int \sqrt{g} d^4x \, \sqrt{g} \left(-\frac{1}{2} g_{\mu\nu} F(R) \delta g^{\mu\nu} + F'(R) \delta R \right) \quad . \tag{20.21}$$

Using (20.19) in the form

$$\delta R = R_{\mu\nu} \delta g^{\mu\nu} - (\nabla_\mu \nabla_\nu - g_{\mu\nu} \Box) \delta g^{\mu\nu} \quad , \tag{20.22}$$

and assuming there are no (or ignoring) boundary terms, so that we can integrate by parts the differential operator acting on $\delta g^{\mu\nu}$ and let it act on $F'(R)$ instead, one finds

$$\delta S = \int \sqrt{g} \left(-\frac{1}{2} g_{\mu\nu} F(R) + F'(R) R_{\mu\nu} - (\nabla_\mu \nabla_\nu - g_{\mu\nu} \Box) F'(R) \right) \delta g^{\mu\nu} \quad . \tag{20.23}$$

From this one can immediately read off the vacuum field equations.

One evident consequence of this is that for non-pathological choices of $F(R)$, a solution of the vacuum Einstein equations ($R_{\mu\nu} = 0, R = 0$, such as the Schwarzschild solution) will continue to be a solution of this $F(R)$-gravity theory, so that such proposed modifications of the Einstein-Hilbert action do not immediately run afoul of precision solar system tests of general relativity.

20.3 Matter Action and the Covariant Energy-Momentum Tensor

In order to obtain the non-vacuum Einstein equations, we need to decide what the matter Lagrangian should be. Now there is an obvious choice for this. If we have matter, then in addition to the Einstein equations we also want the equations of motion for the matter fields. Therefore we should add to the Einstein-Hilbert action the standard minimally coupled matter action

$$S_M[\phi, g_{\alpha\beta}] = \int \sqrt{g} d^4x \; L_M(\phi(x), \partial_\lambda \phi(x), \ldots; g_{\mu\nu}(x), \partial_\lambda g_{\mu\nu}(x), \ldots) \quad , \tag{20.24}$$

ϕ representing any kind of (scalar, vector, tensor, ...) matter field, obtained by suitable covariantisation of the corresponding matter action in Minkowski space via the principle of minimal coupling (section 5). Thus e.g. the matter action for a Klein-Gordon field would be (6.11),

$$S_M[\phi, g_{\alpha\beta}] = \int \sqrt{g} d^4x \left[-\tfrac{1}{2} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi - \tfrac{1}{2} m^2 \phi^2 \right] \quad , \tag{20.25}$$

and that for Maxwell theory would be (6.51),

$$S[A_\alpha, g_{\alpha\beta}] = -\tfrac{1}{4} \int \sqrt{g} d^4x \; g^{\mu\lambda} g^{\nu\rho} F_{\mu\nu} F_{\lambda\rho} \quad . \tag{20.26}$$

Of course, the variation of the matter action with respect to the matter fields will give rise to the covariant equations of motion of the matter fields. If we now want to derive the coupled gravity-matter equations from a variational principle, then the matter contribution to the gravitational field equations (i.e. the source terms for the gravitational field) will necessarily be given by the metric variation of the matter action. As already discussed in detail in section 7.6, we may as well simply define the *covariant energy-momentum tensor* to be the source of the gravitational field equations (7.105),

$$\delta_{\text{metric}} S_M[\phi, g_{\alpha\beta}] = -\tfrac{1}{2} \int \sqrt{g} d^4x \, T_{\mu\nu} \delta g^{\mu\nu} \quad \Leftrightarrow \quad T_{\mu\nu} := -\frac{2}{\sqrt{g}} \frac{\delta}{\delta g^{\mu\nu}} S_M[\phi, g_{\alpha\beta}] \ , \tag{20.27}$$

In particular, we had already seen in section 7.6 that this definition reproduces the known results in the case of a scalar field or Maxwell theory, and that in general it automatically gives a symmetric and gauge invariant tensor without the need for some improvement procedure. It is also automatically covariantly conserved on-shell as a consequence of general covariance of the matter action (cf. section 20.6).

Therefore, the complete gravity-matter action for General Relativity is

$$S[g_{\alpha\beta}, \phi] = \frac{1}{16\pi G_N} S_{EH}[g_{\alpha\beta}] + S_M[\phi, g_{\alpha\beta}] \tag{20.28}$$

with

$$\frac{\delta S[g_{\alpha\beta}, \phi]}{\delta g^{\mu\nu}} = 0 \quad \Leftrightarrow \quad G_{\mu\nu} = 8\pi G_N T_{\mu\nu} \ . \tag{20.29}$$

REMARKS:

1. If one were to try to deduce the gravitational field equations by starting from a variational principle, i.e. by constructing the simplest generally covariant action for the metric and the matter fields (and this would be the modern approach to the problem, had Einstein not already solved it for us a 100 years ago), then one would also invariably be led to the above action.

   The relative numerical factor $16\pi G_N$ between the two terms would of course then not be fixed a priori, because this approach will not (and cannot possibly be expected to) determine Newton's constant. The prefactor could once again be determined by looking at the Newtonian limit of the resulting equations of motion.

2. As we saw above, a cosmological constant term can be included by adding a constant term to the Einstein-Hilbert Lagrangian. One can equally well add a constant term to the matter Lagrangian instead (and this clearly reveals its interpretation as a constant shift of the energy, e.g. by a vacuum energy contribution, of the matter fields).

As we have seen above, the variation of the Einstein-Hilbert action leads to a boundary term that depends not just on the metric but also on the derivatives of the metric. This is related to the fact that the action itself depends also on the second derivatives of the metric. Indeed, it follows from the explicit expression for the scalar curvature in terms of the Christoffel symbols and the metric, obtained by contracting (8.5),

$$R = g^{\mu\nu}(\partial_\lambda \Gamma^\lambda{}_{\mu\nu} - \partial_\nu \Gamma^\lambda{}_{\mu\lambda} + \Gamma^\lambda{}_{\lambda\rho}\Gamma^\rho{}_{\nu\mu} - \Gamma^\lambda{}_{\nu\rho}\Gamma^\rho{}_{\lambda\mu}) \ , \tag{20.30}$$

that the Einstein-Hilbert action contains terms that are quadratic in first derivatives of the metric, as well as terms that depend linearly on the second derivatives.

In ordinary Lagrangian field theory, such linear second derivative terms can usually be introduced or eliminated (depending on what one wants to achieve) by the addition of suitable boundary terms to the action. As an example, consider the action of a free scalar field (in Minkowski space, say):

- The standard action is
$$S_0[\phi] = -\tfrac{1}{2}\int \partial^\alpha \phi \partial_\alpha \phi \ . \tag{20.31}$$
When one uses this action, the boundary term arising from the variation of the action will depend on $\delta\phi$ but not on its derivatives,
$$\delta S[\phi] = \int (\Box\phi)\delta\phi - \int \partial_\alpha(\delta\phi\partial^\alpha\phi) = \int (\Box\phi)\delta\phi - \oint_\Sigma d\sigma_\alpha(\delta\phi\partial^\alpha\phi) \ , \tag{20.32}$$
where the boundary of the integration region is denoted by Σ. This is thus the appropriate action for
$$\text{Dirichlet boundary conditions:} \quad (\delta\phi)|_\Sigma = 0 \ . \tag{20.33}$$

- One can also consider the action
$$S_1[\phi] = \tfrac{1}{2}\int \phi\Box\phi \ . \tag{20.34}$$
This action differs from S_0 by a total derivative (or boundary) term,
$$S_1[\phi] = S_0[\phi] + \tfrac{1}{2}\int \partial_\alpha(\phi\partial^\alpha\phi) = S_0[\phi] + \tfrac{1}{2}\oint d\sigma_\alpha(\phi\partial^\alpha\phi) \ . \tag{20.35}$$
It will therefore give rise to the same Euler-Lagrange bulk equations of motion. In this case, however, the boundary term arising from the variation of the action will depend both on $\delta\phi$ and its derivatives,
$$\delta S_1[\phi] = \int (\Box\phi)\delta\phi + \tfrac{1}{2}\oint d\sigma_\alpha(\phi\partial^\alpha\delta\phi - \delta\phi\partial^\alpha\phi) \ . \tag{20.36}$$
There are no obvious (or at least no obviously useful) boundary conditions compatible with this form of the action.

- Another option is the action

$$\begin{aligned} S_2[\phi] &= S_0[\phi] + \int \partial_\alpha(\phi\partial^\alpha\phi) = \tfrac{1}{2}\int \partial^\alpha\phi\partial_\alpha\phi + \int \phi\Box\phi \\ &= 2S_1[\phi] - S_0[\phi] \ . \end{aligned} \tag{20.37}$$

  Its variation is

$$\delta S_2[\phi] = \int (\Box\phi)\delta\phi + \oint d\sigma_\alpha(\phi\partial^\alpha\delta\phi) \ . \tag{20.38}$$

  In this case, the boundary term only depends on the normal derivative of the variation,

$$d\sigma_\alpha(\phi\partial^\alpha\delta\phi) \sim \phi N^\alpha\partial_\alpha\delta\phi \ , \tag{20.39}$$

  and therefore $S_2[\phi]$ is the appropriate choice of action for

$$\text{Neumann boundary conditions:} \quad (N^\alpha\partial_\alpha\delta\phi)|_\Sigma = 0 \ . \tag{20.40}$$

Let us now return to the case at hand, the action for gravity. In this case second derivative terms are required by general covariance of the action (since there is no scalar that can be constructed solely from the metric and its first derivatives). However, the fact that that the second derivatives appear linearly and that $g^{\alpha\beta}\delta R_{\alpha\beta}$ is a total derivative reflect the fact that these second derivatives are spurious in the sense that they can be eliminated by an integration by parts or by adding a suitable boundary or total derivative term, albeit at the expense of general covariance.

In fact, by straightforward manipulation of (20.30), using identities such as

$$\partial_\lambda g^{\mu\nu} = -g^{\mu\alpha}\ \partial_\lambda g_{\alpha\beta}\ g^{\beta\nu} = -(\Gamma^\mu_{\beta\lambda}g^{\beta\nu} + \Gamma^\nu_{\ \beta\lambda}g^{\beta\mu}) \tag{20.41}$$

and $\partial_\lambda(\sqrt{g}) = \sqrt{g}\ \Gamma^\mu_{\ \mu\lambda}$, one finds that the Lagrangian density $\sqrt{g}R$ can be written as

$$\sqrt{g}R = 2\sqrt{g}\ g^{\alpha\beta}(\Gamma^\mu_{\nu\alpha}\Gamma^\nu_{\ \mu\beta} - \Gamma^\mu_{\alpha\beta}\Gamma^\nu_{\ \nu\mu}) + \partial_\lambda(\sqrt{g}\ B^\lambda) \tag{20.42}$$

where

$$B^\lambda = g^{\mu\nu}\Gamma^\lambda_{\mu\nu} - g^{\mu\lambda}\Gamma^\nu_{\ \nu\mu} \ . \tag{20.43}$$

With due care, one can also write the total derivative term as

$$\partial_\lambda(\sqrt{g}\ B^\lambda) = \sqrt{g}\ \nabla_\lambda B^\lambda \ , \tag{20.44}$$

as long as one remembers that B^λ is not a tensor. Either way, we see that instead of the generally covariant Einstein-Hilbert action one can use the non-covariant but quadratic action

$$S_{EH}[g_{\alpha\beta}] \to S_E[g_{\alpha\beta}] = \int \sqrt{g}d^4x\ 2g^{\alpha\beta}(\Gamma^\mu_{\nu\alpha}\Gamma^\nu_{\ \mu\beta} - \Gamma^\mu_{\alpha\beta}\Gamma^\nu_{\ \nu\mu}) \ . \tag{20.45}$$

This action was originally considered by Einstein himself and is therefore also known as the *Einstein action*.

It will be useful for the following to write this in the form

$$S_E[g_{\alpha\beta}] = S_{EH}[g_{\alpha\beta}] - \int \sqrt{g} d^4x \; \nabla_\lambda B^\lambda = S_{EH}[g_{\alpha\beta}] - \oint d\sigma_\lambda B^\lambda \quad , \tag{20.46}$$

in which the non-covariant terms are now manifestly confined to the boundary of the region of integration. This is now a reasonably respectable action, but there is a more attractive variant of this construction which we will discuss in the next section.

20.5 Gibbons-Hawking-York Boundary Term

In section 20.2 we have seen that the metric variation of the Einstein-Hilbert action has the form

$$\delta S_{EH}[g_{\alpha\beta}] = \int \sqrt{g} d^4x \; G_{\alpha\beta} \delta g^{\alpha\beta} + \int \sqrt{g} d^4x \; \nabla_\lambda (\Delta B)^\lambda \quad , \tag{20.47}$$

where

$$\begin{aligned} (\Delta B)^\lambda &= g^{\mu\nu} \delta \Gamma^\lambda_{\mu\nu} - g^{\mu\lambda} \delta \Gamma^\nu_{\nu\mu} \\ &= (g^{\lambda\alpha} g^{\nu\beta} - g^{\lambda\nu} g^{\alpha\beta}) \nabla_\nu \delta g_{\alpha\beta} \quad . \end{aligned} \tag{20.48}$$

The reason for this notation, and the relation between this object $(\Delta B)^\lambda$ and the quantity B^λ introduced in (20.43) will be explained below. The first (bulk) term gives us as the Euler-Lagrange equations the vacuum Einstein equations $G_{\alpha\beta} = 0$, while the second term is a total derivative.

Thus, when one performs the integral over a space-time region $\mathcal{V}$ bounded by the hypersurface $\partial\mathcal{V} = \Sigma$ (which we shall assume to be spacelike or timelike), upon use of the Gauss integral formula (16.47) one finds that the second (total derivative) term can be written as

$$\begin{aligned} \int_{\mathcal{V}} \sqrt{g} d^4x \; \nabla_\lambda (\Delta B)^\lambda &= \oint_\Sigma d\sigma_\lambda (\Delta B)^\lambda \\ &= \epsilon \oint_\Sigma d^n y \; \sqrt{h} \; N_\lambda (\Delta B)^\lambda \quad , \end{aligned} \tag{20.49}$$

where N_λ is the normal vector to the boundary Σ in $\mathcal{V}$ and h_{ab} is the induced metric on the boundary. Explicitly the boundary integrand is

$$N_\lambda (\Delta B)^\lambda = (N^\rho g^{\mu\nu} - N^\mu g^{\rho\nu}) \nabla_\mu \delta g_{\rho\nu} \quad . \tag{20.50}$$

Using the decomposition

$$g^{\mu\nu} = h^{\mu\nu} + \epsilon N^\mu N^\nu \tag{20.51}$$

of the metric on Σ (with $N^\mu N_\mu = \epsilon$), one sees that the terms with 3 N's cancel, and one is left with

$$N_\lambda (\Delta B)^\lambda = N^\rho h^{\mu\nu} \nabla_\mu \delta g_{\rho\nu} - N^\mu h^{\rho\nu} \nabla_\mu \delta g_{\rho\nu} \quad . \tag{20.52}$$

The first term only depends on the variations $\delta g_{\rho\nu}$ on the boundary, and its tangential derivatives $h^{\mu\nu} \nabla_\mu \delta g_{\rho\nu}$. Therefore that term is zero if one imposes standard Dirichlet boundary conditions

$$\delta g_{\alpha\beta}|_\Sigma = 0 \tag{20.53}$$

on the metric at Σ. The second term, on the other hand, depends on $\delta g_{\rho\nu}$ and its normal derivative $N^\mu \nabla_\mu \delta g_{\rho\nu}$, and is therefore non-zero,

$$\delta g_{\alpha\beta}|_\Sigma = 0 \quad \Rightarrow \quad N_\lambda (\Delta B)^\lambda|_\Sigma = -N^\mu h^{\rho\nu} \nabla_\mu \delta g_{\rho\nu}|_\Sigma = -h^{\rho\nu} N^\mu \partial_\mu \delta g_{\rho\nu}|_\Sigma \ . \tag{20.54}$$

Therefore with Dirichlet boundary conditions the variation of the Einstein-Hilbert action gives rise to a non-zero boundary term,

$$\delta S_{EH}[g_{\alpha\beta}] = \int \sqrt{g} d^4x \, G_{\alpha\beta} \delta g^{\alpha\beta} + \epsilon \oint_\Sigma \sqrt{h} d^3y \, (-h^{\rho\nu} N^\mu \partial_\mu \delta g_{\rho\nu}) \ . \tag{20.55}$$

It is therefore not true that the variation of the Einstein-Hilbert action is the Einstein tensor,

$$\frac{\delta}{\delta g^{\alpha\beta}} S_{EH} \neq \sqrt{g} G_{\alpha\beta} \ . \tag{20.56}$$

In fact, the presence of this boundary term means that the functional $S_{EH}[g_{\alpha\beta}]$ is not even differentiable (in the sense of variational calculus).

The way to resolve this issue is, as for the scalar field discussed above, to add a suitable boundary term to the action itself. This will not change the bulk variation, and it turns out that e.g. for Dirichlet boundary conditions the boundary term can be chosen in such a way that its variation cancels the boundary term above.

Actually, we already have one candidate for the boundary term, namely the one relating the Einstein and Einstein-Hilbert actions in (20.46). The variation of the Einstein action is

$$\begin{aligned} \delta S_E[g_{\alpha\beta}] &= \delta S_{EH}[g_{\alpha\beta}] - \delta \oint d\sigma_\lambda B^\lambda \\ &= \int \sqrt{g} d^4x \, G_{\alpha\beta} \delta g^{\alpha\beta} + \oint_\Sigma d^n y \, \sqrt{h} \, \epsilon N_\lambda (\Delta B)^\lambda - \delta \Big(\oint_\Sigma d^n y \, \sqrt{h} \, \epsilon N_\lambda B^\lambda \Big) \ . \end{aligned} \tag{20.57}$$

Calculating the variation in the second term for Dirichlet boundary conditions on Σ, one finds

$$\delta \Big(\oint_\Sigma d^n y \, \sqrt{h} \, \epsilon N_\lambda B^\lambda \Big) = \oint_\Sigma d^n y \, \sqrt{h} \, \epsilon N_\lambda \delta B^\lambda \ , \tag{20.58}$$

with

$$\delta B^\lambda = \delta (g^{\mu\nu} \Gamma^\lambda_{\mu\nu} - g^{\mu\lambda} \Gamma^\nu{}_{\nu\mu}) \quad = g^{\mu\nu} \delta \Gamma^\lambda_{\mu\nu} - g^{\mu\lambda} \delta \Gamma^\nu{}_{\nu\mu} = (\Delta B)^\lambda \ . \tag{20.59}$$

Thus for Dirichlet boundary conditions, the variation of B^λ on the boundary equals the quantity $(\Delta B)^\lambda$ (20.48) arising from the variation of the Einstein-Hilbert action. As a consequence, there is no boundary term in the variation of the Einstein action for Dirichlet boundary conditions,

$$\delta S_E[g_{\alpha\beta}] = \int \sqrt{g} d^4x \, G_{\alpha\beta} \delta g^{\alpha\beta} \tag{20.60}$$

and the Einstein action leads to a well-defined variational principle (with a differentiable action).

Although this is progress, the boundary term that one adds to the Einstein-Hilbert action to obtain the Einstein action is not particularly attractive, in particular as it is non-covariant not only with respect to bulk coordinate transformations but also with respect to boundary coordinate transformations (i.e. the integrand is not a Σ-scalar in the terminology of section 15.1).

This is something one would have to live with if one could not do better. However, such a boundary term achieving this is not unique as it is evidently only defined up to terms whose variations vanish for Dirichlet boundary conditions, in particular up to terms that only depend on the intrinsic geometry of Σ. Among these candidates there is a preferred, geometrically transparent, boundary term, the *Gibbons-Hawking-York* boundary term,

$$S_{GHY}[g_{\alpha\beta}] = 2\epsilon \oint_\Sigma \sqrt{h} d^3y \; K \quad . \tag{20.61}$$

Here

$$\begin{aligned} K &= h^{ab}K_{ab} = h^{\alpha\beta}\nabla_\alpha N_\beta \\ &= (g^{\alpha\beta} + N^\alpha N^\beta)\nabla_\alpha N_\beta \\ &= g^{\alpha\beta}\nabla_\alpha N_\beta = \nabla_\alpha N^\alpha \end{aligned} \tag{20.62}$$

is the trace of the extrinsic curvature K_{ab} of Σ (discussed in section 18).

One way to prove that this is a good boundary term is to determine its variation and to show that it cancels against the variation arising from the Einstein-Hilbert action for Dirichlet boundary conditions. Alternatively one should show that the above boundary term differs from the boundary term for the Einstein action only by expressions that depend on the metric $g_{\alpha\beta}$ and its tangential derivatives.

Since usually one does the former, let us do the latter. The difference between the two boundary integrands is

$$2K + N_\lambda B^\lambda = 2\nabla_\lambda N^\lambda + N_\lambda B^\lambda \quad . \tag{20.63}$$

A calculation identical to the one leading to $N_\lambda(\Delta B)^\lambda$ in (20.52) shows that

$$N_\lambda B^\lambda = N^\rho h^{\mu\nu}\partial_\mu g_{\rho\nu} - N^\mu h^{\rho\nu}\partial_\mu g_{\rho\nu} \quad . \tag{20.64}$$

Here the first term only depends on the metric and its tangential derivatives, while the second term involves normal derivatives of the metric. On the other hand, for K we have

$$2K = h^{\nu\rho}L_N g_{\nu\rho} = h^{\nu\rho}N^\mu\partial_\mu g_{\nu\rho} + 2h^\mu_{\ \nu}\partial_\mu N^\nu \quad , \tag{20.65}$$

where we have used the non-covariant way (9.39) of writing the Lie derivative,

$$L_N g_{\mu\nu} = N^\lambda\partial_\lambda g_{\mu\nu} + \partial_\mu N^\lambda g_{\lambda\nu} + \partial_\nu N^\lambda g_{\mu\lambda} \quad . \tag{20.66}$$

Thus the first term cancels the normal derivative term in (20.64) and the remaining terms in (20.64) and (20.65) only involve fields $g_{\alpha\beta}$, i.e. $h_{\alpha\beta}$ and N_α, and their tangential derivatives that are fixed on the boundary.

This establishes that the Gibbons-Hawking-York boundary term is an acceptable choice of boundary term. This is also the standard choice, and gives rise to the standard gravitational action

$$\begin{aligned} S_g[g_{\alpha\beta}] &= S_{EH}[g_{\alpha\beta}] + S_{GHY}[g_{\alpha\beta}] \\ &= \int \sqrt{g} d^4x \; R + 2\epsilon \oint_\Sigma \sqrt{h} d^3y \; K \quad , \end{aligned} \tag{20.67}$$

expressed purely in terms of the intrinsic and extrinsic scalar curvatures R and K, with

$$\delta S_g[g_{\alpha\beta}] = \int \sqrt{g} d^4x \; G_{\alpha\beta} \delta g^{\alpha\beta} \tag{20.68}$$

for Dirichlet boundary conditions.

Remarks:

1. In addition, one can add terms to the action that do not depend on the dynamical fields (as this will certainly change neither the variation with respect to the dynamical fields nor the equations of motion). A common choice is a kind of "background subtraction", designed to associate the numerical value $S = 0$ to a particular background metric $g^0_{\alpha\beta}$.[38] Thus one could define the "physical" action to be

$$S[g_{\alpha\beta}] = S_g[g_{\alpha\beta}] - S_g[g^0_{\alpha\beta}] \quad . \tag{20.69}$$

   In particular, if one is interested in asymptotically flat space-times, say, then the appropriate reference background metric is just the flat Minkowski metric, and then (20.69) takes the simple form

$$S[g_{\alpha\beta}] = \int \sqrt{g} d^4x \; R + 2\epsilon \oint_\Sigma \sqrt{h} d^3y \; (K - K^0) \quad , \tag{20.70}$$

   where K^0 is the trace of the extrinsic curvature of the boundary (isometrically) embedded into Minkowski space. In section 21.12 we will be led to a similar subtraction prescription at the level of the Hamiltonian.

2. Another way to motivate (or arrive at) the Gibbons-Hawking-York boundary term is to start from the decomposition (18.73)

$$R = \bar{R} + \epsilon(K^2 - K^{\alpha\beta}K_{\alpha\beta}) + 2\epsilon\nabla_\alpha(N^\beta \nabla_\beta N^\alpha - N^\alpha \nabla_\beta N^\beta) \quad . \tag{20.71}$$

   of the Ricci scalar provided by the Gauss-Codazzi equations. This turns out to be a convenient starting point for the canonical (Hamiltonian) formulation of general relativity, and we will therefore discuss this in that context in section 21.2.

[38] See e.g. S. Hawking, G. Horowitz, *The Gravitational Hamiltonian, Action, Entropy, and Surface Terms*, `arXiv:gr-qc/9501014`, for a brief discussion.

20.6 General Covariance and Noether Identities

The variational (i.e. action or Lagrangian based) formulation of general relativity has a number of significant conceptual and technical advantages, and we will explore some of them in this section.

- I mentioned before, in section 19.7, that it is no accident that the Bianchi identities come to the rescue of the general covariance of the Einstein equations in the sense that they reduce the number of independent equations from ten to six. We will now see that indeed the Bianchi identities are a consequence of the general covariance of the Einstein-Hilbert action.
- Virtually the same calculation will show that the covariant (metric, Hilbert) energy-momentum tensor, as defined above, is automatically conserved (on shell) by virtue of the general covariance of the matter action.
- This argument can also be turned around to show that (generically, at least) the Einstein equations imply the matter equations of motion, a very characteristic feature of generally covariant gravitational field equations.
- Simple variants of these arguments will also provide us with the Noether currents associated with the general covariance of the Einstein-Hilbert action.
- Analogous considerations, but now applied to the minimally coupled generally covariant matter action, will provide us with some insight into the relation between the (Belinfante improved) canonical and covariant energy-momentum tensors introduced in section 7.

To set the stage, we need to discuss how to express general covariance of an action, either of the Einstein-Hilbert gravitational action $S_{EH}[g_{\alpha\beta}]$ or of some (minimally) gravitationally coupled general covariant matter action $S_M[\phi, g_{\alpha\beta}]$, in a form that allows us to explore its consequences in a Lagrangian formalism.

At first, the statement that an action is generally covariant means that it is invariant under transformations $x^\alpha \to x'^\alpha$ of the coordinates (for present purposes it will for once be more convenient to use the same indices on the old and new coordinates and to distinguish transformed objects by primes), and the accompanying tensorial transformation of the fields, given e.g. by

$$\begin{aligned} x \to x' \quad \Rightarrow \quad & \phi(x) \to \phi'(x') = \phi(x) \\ & g_{\alpha\beta}(x) \to g'_{\alpha\beta}(x') = \frac{\partial x^\gamma}{\partial x'^\alpha} \frac{\partial x^\delta}{\partial x'^\beta}\, g_{\gamma\delta}(x) \quad . \end{aligned} \tag{20.72}$$

So far, this is utterly familiar, but since in the action one is integrating over the coordinates x^α, say, we would like to express this invariance not as a statement between

transformed fields at x' and the old fields at x but in terms of transformations of the fields at a point x (which we can then plug into the Lagrangian or action). This means that we do not want to consider the transformation $\phi(x) \to \phi'(x')$ (and its counterpart for the metric) but rather the transformations

$$\phi(x) \to \phi'(x) \quad , \quad g_{\alpha\beta}(x) \to g'_{\alpha\beta}(x) \ . \tag{20.73}$$

In particular, for infinitesimal coordinate transformations of the form

$$x'^{\alpha} = x^{\alpha} + \epsilon\xi^{\alpha}(x) \tag{20.74}$$

or (suppressing the ϵ)

$$\delta_{\xi} x^{\alpha} = \xi^{\alpha}(x) \ , \tag{20.75}$$

the infinitesimal variations of the fields are then precisely the Lie derivatives of the fields discussed in section 9,

$$\delta_{\xi}\phi(x) = L_{\xi}\phi(x) = \xi^{\alpha}\partial_{\alpha}\phi(x) \tag{20.76}$$

for a scalar field,

$$\delta_{\xi} g_{\alpha\beta}(x) = L_{\xi} g_{\alpha\beta}(x) = \nabla_{\alpha}\xi_{\beta}(x) + \nabla_{\beta}\xi_{\alpha}(x) \ , \tag{20.77}$$

for the metric, etc.

As a reminder, a quick way to derive this transformation of the metric is to start with the tensorial transformation behaviour in the form

$$(g_{\alpha\beta}(x') - g'_{\alpha\beta}(x'))dx'^{\alpha}dx'^{\beta} = g_{\alpha\beta}(x')dx'^{\alpha}dx'^{\beta} - g_{\alpha\beta}(x)dx^{\alpha}dx^{\beta} \ , \tag{20.78}$$

and to then apply this to the infinitesimal transformation (20.74). Expanding the differentials

$$dx'^{\alpha} = dx^{\alpha} + \epsilon(\partial_{\gamma}\xi^{\alpha})dx^{\gamma} \tag{20.79}$$

and the components $g_{\alpha\beta}(x')$ of the metric to first order in ϵ,

$$g_{\alpha\beta}(x') = g_{\alpha\beta}(x) + \epsilon\xi^{\gamma}\partial_{\gamma}g_{\alpha\beta}(x) \ , \tag{20.80}$$

one finds (20.77) (in its not manifestly covariant form (9.39)).

To see that this indeed leads to a symmetry for any generally covariant action, i.e. any action of the form

$$S = \int \sqrt{g}d^4x \ L(x) \tag{20.81}$$

where L is a scalar, note that for any density $\sqrt{g}F$, F a scalar, one has, by the by now familiar identity for the variation of $\sqrt{g}$,

$$\begin{aligned} \delta_{\xi}(\sqrt{g}F) &= (\delta_{\xi}\sqrt{g})F + \sqrt{g}\delta_{\xi}F = \tfrac{1}{2}\sqrt{g}g^{\alpha\beta}(L_{\xi}g_{\alpha\beta})F + \sqrt{g}L_{\xi}F \\ &= \sqrt{g}(\nabla_{\alpha}\xi^{\alpha})F + \sqrt{g}\xi^{\alpha}\partial_{\alpha}F = \partial_{\alpha}(\sqrt{g}\ \xi^{\alpha}F) \ , \end{aligned} \tag{20.82}$$

i.e.

$$\delta_\xi(\sqrt{g}F) = \partial_\alpha(\sqrt{g}\,\xi^\alpha F) = \sqrt{g}\,\nabla_\alpha(\xi^\alpha F) \ , \tag{20.83}$$

a result previously obtained in (9.69). Thus the variation of a generally covariant action,

$$\delta_\xi S = \int \sqrt{g}d^4x\,\nabla_\alpha(\xi^\alpha L) \ , \tag{20.84}$$

is a total derivative and $\delta_\xi = L_\xi$ certainly generates a symmetry in the usual Lagrangian / Noether sense. On the other hand, $\delta_\xi(\sqrt{g}L)$ can also be expressed in terms of the δ_ξ-variations of the fields, and combining this with the invariance of the action (up to boundary terms), one can now deduce the consequences of general covariance for a given theory, either for ξ that vanish at the boundary of the integration region or, more generally, for ξ that are non-trivial there. We will now discuss these possibilities in turn for the Einstein-Hilbert and matter actions.

1. Contracted Bianchi Identities

   As our first application we consider the Einstein-Hilbert action, and its associated invariance under all infinitesimal coordinate transformations generated by ξ that vanish on the boundary. In that case, from the considerations above we know that the Einstein-Hilbert action is strictly invariant,

   $$\delta_\xi S_{EH} = 0 \ . \tag{20.85}$$

   On the other hand we also know that, modulo boundary terms,

   $$\delta S_{EH} = \int \sqrt{g}d^4x\,G_{\alpha\beta}\delta g^{\alpha\beta} = -\int \sqrt{g}d^4x\,G^{\alpha\beta}\delta g_{\alpha\beta} \tag{20.86}$$

   for any metric variation. Combining these two facts we arrive at the conclusion that

   $$\begin{aligned} 0 = \delta_\xi S_{EH} &= -\int \sqrt{g}d^4x\,G^{\alpha\beta}\delta_\xi g_{\alpha\beta} \\ &= -2\int \sqrt{g}d^4x\,G^{\alpha\beta}\nabla_\alpha\xi_\beta \\ &= +2\int \sqrt{g}d^4x\,(\nabla_\alpha G^{\alpha\beta})\xi_\beta \ . \end{aligned} \tag{20.87}$$

   Since this has to hold for all ξ (vanishing on the boundary), we deduce

   $$\delta_\xi S_{EH} = 0 \quad \forall\,\xi \quad \Rightarrow \quad \nabla^\alpha G_{\alpha\beta} = 0 \ . \tag{20.88}$$

   As promised and anticipated, the contracted Bianchi identities are a consequence of the general covariance of the Einstein-Hilbert action.

   To further strengthen the analogy with the gauge invariance of Maxwell theory emphasised in the discussion in section 19.7, note that also the relevant "contracted Bianchi" identity $\partial_a\partial_b F^{ab} = 0$ can be derived from gauge invariance of the Maxwell

action (although this is of course not the most economical way of arriving at this identity in the case at hand).

Indeed, the general variation of the Maxwell action (for variations vanishing on the boundary) is

$$\delta S[A] = \int d^4x (\partial_a F^{ab}) \delta A_b \ . \tag{20.89}$$

The action is invariant under gauge transformations

$$\delta_\Psi A_a = \partial_a \Psi \ , \tag{20.90}$$

i.e. one has

$$0 = \delta_\Psi S[A] = \int d^4x (\partial_a F^{ab}) \partial_b \Psi = \int d^4x (\partial_a \partial_b F^{ab}) \Psi \tag{20.91}$$

(for any Ψ vanishing at the boundary), and thus $\partial_a \partial_b F^{ab} = 0$.

2. Identically Conserved Noether Currents

The fact that one obtains kinematical identities rather than non-trivially conserved currents (non-trivial in the sense that their conservation requires the validity of some dynamical equations of motion) is a characteristic feature of Noether's theorem applied to local (gauge) symmetries. We can also see this when we consider general vector fields ξ (not constrained by the requirement that they give rise to vanishing boundary terms for the given integration domain). In that case, $\delta_\xi S_{EH}$ will not be identically zero but will be a total derivative, and also the corresponding Noether currents will turn out to be identically conserved.

Thus we now consider again the Einstein-Hilbert action, but now with ξs that are allowed to be non-zero on the boundary,

$$\delta_\xi S_{EH} = \int \sqrt{g} d^4x \ \nabla_\alpha (\xi^\alpha R) \ . \tag{20.92}$$

By explicitly performing this variation, as above, the bulk (Einstein tensor) term is identically zero by the contracted Bianchi identity, but we obtain one total derivative term from (20.19),

$$\begin{aligned} g^{\alpha\beta} \delta_\xi R_{\alpha\beta} &= \nabla_\mu [(g^{\mu\alpha} g^{\nu\beta} - g^{\mu\nu} g^{\alpha\beta}) \nabla_\nu L_\xi g_{\alpha\beta}] \\ &= \nabla_\mu [(g^{\mu\alpha} g^{\nu\beta} - g^{\mu\nu} g^{\alpha\beta}) \nabla_\nu (\nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha)] \ , \end{aligned} \tag{20.93}$$

and a second total derivative term

$$\nabla_\mu (-2 G^\mu_{\ \nu} \xi^\nu) = \nabla_\mu (-2 R^\mu_{\ \nu} \xi^\nu + \xi^\mu R) \tag{20.94}$$

from the integration by parts performed in the course of the calculation in (20.87). The term involving the scalar curvature is identical to, and cancels against, the

scalar curvature term arising from (20.92). One is thus left with the statement that for any vector field ξ^μ and any integration domain one has

$$\int \sqrt{g} d^4x \nabla_\mu [-2R^\mu_\nu \xi^\nu + (g^{\mu\alpha} g^{\nu\beta} - g^{\mu\nu} g^{\alpha\beta}) \nabla_\nu (\nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha)] = 0 \ . \qquad (20.95)$$

Thus for any ξ^μ one has the conserved Noether current

$$J^\mu(\xi) = R^\mu_\nu \xi^\nu + \tfrac{1}{2}(g^{\mu\nu} g^{\alpha\beta} - g^{\mu\alpha} g^{\nu\beta}) \nabla_\nu (\nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha) \ . \qquad (20.96)$$

This is precisely the identically conserved current (13.51), (13.53)

$$J^\mu(\xi) = \nabla_\nu (\nabla^{[\mu} \xi^{\nu]}) \quad \Rightarrow \quad \nabla_\mu J^\mu(\xi) = 0 \qquad (20.97)$$

already mentioned in section 13.7. We thus learn that the generalised Komar currents of that section are indeed, as anticipated there, precisely the identically conserved Noether currents associated to the general covariance of the Einstein-Hilbert action.

Note that, as mentioned above, it is a general feature of Noether currents associated to local (gauge) symmetries that they are in fact identically conserved: the current $J^\mu(\xi)$ (or its conterpart for some local symmetry of another theory) cannot possibly be conserved for all possible $\xi^\mu(x)$ unless it is actually identically conserved.[39] In particular, this implies that the conserved charges associated with these currents can always be expressed as surface integrals.

3. On-Shell Covariant Conservation of the Energy-Momentum Tensor

Now let us play the same game with the matter action S_M (20.24). Once again, the variation $\delta_\xi S_M$, expressed in terms of the Lie derivatives $L_\xi g_{\mu\nu}$ and $\delta_\xi \phi = L_\xi \phi$ of the matter fields should be identically zero, by general covariance of the matter action (for the time being we again at first only consider ξ which are such that any boundary terms vanish). Proceeding as before, and using the definition (7.105) of the energy-momentum tensor, we find

$$\begin{aligned} 0 &= \delta_\xi S_M \\ &= \int \sqrt{g} d^4x \, (-\tfrac{1}{2} T_{\mu\nu} \delta_\xi g^{\mu\nu} + \frac{\delta L_M}{\delta \phi} \delta_\xi \phi) \\ &= -\int \sqrt{g} d^4x \, (\nabla^\mu T_{\mu\nu}) \xi^\nu + \int \sqrt{g} d^4x \, \frac{\delta L_M}{\delta \phi} \delta_\xi \phi \ . \end{aligned} \qquad (20.98)$$

[39] See e.g. B. Julia, S. Silva, *Currents and Superpotentials in classical gauge invariant theories I: Local results with applications to Perfect Fluids and General Relativity*, `arXiv:gr-qc/9804029` for a rather explicit elementary argument, and R. Wald, *On identically closed forms locally constructed from a field*, J. Math. Phys. 31 (1990) 2378, R. Wald, *Black Hole Entropy is Noether Charge*, `arXiv:gr-qc/9307038`, V. Iyer and R. Wald, *Some properties of Noether charge and a proposal for dynamical black hole entropy*, `arXiv:gr-qc/9403028` and references therein for related considerations and further developments.

Now once again this has to hold for all ξ, and as the second term is identically zero 'on-shell', i.e. for ϕ satisfying the matter Euler-Lagrange equations of motion $\delta L_M/\delta\phi = 0$, we deduce that

$$\delta_\xi S_M = 0 \quad \forall\, \xi \quad \Rightarrow \quad \nabla^\mu T_{\mu\nu} = 0 \qquad \text{on-shell} \ . \tag{20.99}$$

This should be contrasted with the contracted Bianchi identities which are valid 'off-shell'. The more general situation with the ξ not restricted to vanish on the boundary will be analysed in detail in section 22.2.

4. Einstein Equations Imply (generically) the Matter Equations of Motion

   Note that, to a certain extent, this argument can also be turned around to show that *the equations of motion of the gravitational field generated by some matter fields imply the equations of motion of these matter fields*!

   Indeed, we already know that the Einstein equations imply covariant conservation of the energy-momentum tensor,

$$G_{\alpha\beta} = 8\pi G_N\, T_{\alpha\beta} \quad \Rightarrow \quad \nabla_\alpha T^{\alpha\beta} = 0 \ . \tag{20.100}$$

   Hence, by the above reasoning, the Einstein equations imply

$$G_{\alpha\beta} = 8\pi G_N\, T_{\alpha\beta} \quad \Rightarrow \quad \int \sqrt{g} d^4x\, \frac{\delta L_M}{\delta\phi}\delta_\xi\phi = 0 \tag{20.101}$$

   for all compactly supported ξ. Excluding certain non-generic cases (like for example a constant scalar field for which $\delta_\xi\phi = 0$), one sees that

$$G_{\alpha\beta} = 8\pi G_N\, T_{\alpha\beta} \quad \overset{\text{generically}}{\Longrightarrow} \quad \frac{\delta L_M}{\delta\phi} = 0 \ . \tag{20.102}$$

   This should be contrasted with the Maxwell equations in the presence of (charged) matter fields, say, which only imply current conservation,

$$\partial_\mu F^{\mu\nu} \sim j^\nu \quad \Rightarrow \quad \partial_\nu j^\nu = 0 \ , \tag{20.103}$$

   but not the complete equations of motion of the matter fields.

   To explain what is special about the generally covariant gravitational field equations in this respect, I will conclude this section with a quote by Misner, Thorne and Wheeler, since I could not possibly state this more eloquently:

> The Maxwell equations are so constructed that they automatically fulfill and demand the conservation of charge; but not everything has charge. The Einstein field equation is so constructed that it automatically fulfills and demands the conservation of momentum-energy; and everything does have energy. The Maxwell field equations are indifferent to the interposition of an "external" force, because that force in no way threatens

the principle of conservation of charge. The Einstein field equation cares about every force, because every force is a medium for the exchange of energy.

Electromagnetism has the motto, "I count all the electric charge that's here." All that bears no charge escapes its gaze.

"I weigh all that's here" is the motto of spacetime curvature. No physical entity escapes this surveillance.[40]

20.7 First Order Form of the Action, Torsion and the Palatini Principle

For certain purposes (e.g. as a precursor to a Hamiltonian formulation) it can be useful to put an action leading to 2nd order differential equations into "1st order form" by the introduction of some auxiliarly variables. The prototypical example is Maxwell theory, whose original action (cf. section 6.5)

$$S[A] = -\tfrac{1}{4}\int d^4\xi \; F_{ab}F^{ab} \tag{20.104}$$

depends quadratically on the 1st derivatives of the gauge field A_a, and leads to the 2nd order equations of motion

$$\partial_a F^{ab} = 0 \ . \tag{20.105}$$

To put this into 1st order form, one treats A_a and $F_{ab} \to \mathcal{F}_{ab}$ as a priori *independent* fields and considers the action

$$\begin{aligned} S[A,\mathcal{F}] &= \int d^4\xi \ \left(-(\partial_a A_b)\mathcal{F}^{ab} + \tfrac{1}{4}\mathcal{F}_{ab}\mathcal{F}^{ab}\right) \\ &= \int d^4\xi \ \left(-\tfrac{1}{2}(\partial_a A_b - \partial_b A_a)\mathcal{F}^{ab} + \tfrac{1}{4}\mathcal{F}_{ab}\mathcal{F}^{ab}\right) \ , \end{aligned} \tag{20.106}$$

which depends purely algebraically on $\mathcal{F}_{ab}$ and only linearly on the 1st derivatives of A_a. The equations of motion arising from the variation of A_a are the 1st order equations

$$\delta A \quad \Rightarrow \quad \partial_a \mathcal{F}^{ab} = 0 \ . \tag{20.107}$$

However these are not (yet) the vacuum Maxwell equations because $\mathcal{F}_{ab}$ is not (yet) to be identified with the Maxwell field stength tensor of A_a. This identification now results from the (purely algebraic) equation of motion associated with variations of $\mathcal{F}_{ab}$,

$$\delta\mathcal{F} \quad \Rightarrow \quad \mathcal{F}_{ab} = \partial_a A_b - \partial_b A_a = F_{ab} \ . \tag{20.108}$$

Plugging this result into the previous equations then gives rise to the standard Maxwell equations (and plugging it into the 1st order action $S[A,\mathcal{F}]$ reduces it to the standard Maxwell action $S[A]$).

[40] C. Misner, K. Thorne, J. Wheeler, *Gravitation*, section 20.6.

Something similar (but more interesting and somewhat more subtle) can be done in the case of general relativity.

Recall from sections 5.4 and 11.5 that a priori a metric $g_{\mu\nu}$ and a connection $\tilde{\Gamma}^{\lambda}_{\ \mu\nu}$ are independent concepts, and that the notion of curvature (curvature and Ricci tensors) can be defined for an arbitrary connection,

$$R^{\lambda}_{\ \sigma\mu\nu}(\tilde{\Gamma}) = \partial_\mu \tilde{\Gamma}^{\lambda}_{\ \sigma\nu} - \partial_\nu \tilde{\Gamma}^{\lambda}_{\ \sigma\mu} + \tilde{\Gamma}^{\lambda}_{\ \mu\rho}\tilde{\Gamma}^{\rho}_{\ \nu\sigma} - \tilde{\Gamma}^{\lambda}_{\ \nu\rho}\tilde{\Gamma}^{\rho}_{\ \mu\sigma} \quad , \quad R_{\mu\nu}(\tilde{\Gamma}) = R^{\lambda}_{\ \mu\lambda\nu}(\tilde{\Gamma}) \ . \tag{20.109}$$

General relativity employs and is formulated in terms of the canonical Levi-Civita connection described by the Christoffel symbols $\tilde{\Gamma}^{\lambda}_{\ \mu\nu} = \Gamma^{\lambda}_{\ \mu\nu}$, characterised by the fact that the connection is compatible with the metric and has no torsion. It is thus easy to come up with various generalisations of general relativity in which these requirements are relaxed. We will not get into these matters here.[41]

However, it is a curious, and occasionally calculationally or conceptually useful, fact that it is possible to relax somewhat the a priori identification of the connection with the Levi-Civita connection and nevertheless reproduce general relativity by treating the connection and metric as independent variables.

Specifically, we will consider an action of the generalised Einstein-Hilbert-like form

$$S[g_{\mu\nu}, \tilde{\Gamma}^{\lambda}_{\ \mu\nu}] = \int \sqrt{g} d^4x \ R(\tilde{\Gamma}) = \int \sqrt{g} d^4x \ g^{\mu\nu} R_{\mu\nu}(\tilde{\Gamma}) \tag{20.110}$$

for a (yet to be specified) class of connections $\tilde{\Gamma}^{\lambda}_{\ \mu\nu}$, with $g_{\mu\nu}$ and $\tilde{\Gamma}^{\lambda}_{\ \mu\nu}$ to be treated as independent variables. Since $R_{\mu\nu}(\tilde{\Gamma})$ does then not depend on the metric, the action depends purely algebraically on the metric, and on at most 1st derivatives of the connection (linearly!).

One key simplification of this kind of action is that the variation with respect to the metric is elementary (and identical to the variation of the $\sqrt{g}g^{\mu\nu}$ terms of the Einstein-Hilbert Lagrangian density $\sqrt{g}g^{\mu\nu}R_{\mu\nu}$), namely

$$\delta_g S[g_{\mu\nu}, \tilde{\Gamma}^{\lambda}_{\ \mu\nu}] = \int \sqrt{g} d^4x \ \left(R_{\mu\nu}(\tilde{\Gamma}) - \tfrac{1}{2} g_{\mu\nu} R(\tilde{\Gamma}) \right) \delta g^{\mu\nu} \tag{20.111}$$

(no integration by parts or identification of total derivative terms required). Thus, in the absence of the coupling of the metric (or of gravity) to other fields we find the equations of motion

$$G_{\mu\nu}(\tilde{\Gamma}) \equiv R_{\mu\nu}(\tilde{\Gamma}) - \tfrac{1}{2} g_{\mu\nu} R(\tilde{\Gamma}) = 0 \ . \tag{20.112}$$

These are, however, not yet the vacuum Einstein equations because the independent connection $\tilde{\Gamma}$ is not the Levi-Civita connection.

[41] Many of these generalisations (including theories with non-symmetric metrics) were originally explored by Einstein and his collaborators in their futile and (at least by the 1930s) ill-motivated attempts to find a unified field theory of gravity and Maxwell theory. For details, see e.g. the review H. Goenner, *On the History of Unified Field Theories*, Living Rev. Relativity 7 (2004) 2, `http://www.livingreviews.org/lrr-2004-2`.

It remains to look at the equations of motion imposed by stationarity of the action with respect to variations of $\tilde{\Gamma}$. It turns out (the *Palatini principle*) that

1. if one chooses the connections to be torsion-free and imposes the $\tilde{\Gamma}$-equations of motion, then the connections are forced to also be compatible with the metric and thus $\tilde{\Gamma}$ is uniquely determined to be the Levi-Civita connection
2. if one chooses the connections to be compatible with the metric and imposes the $\tilde{\Gamma}$-equations of motion, then the connections are forced to also be torsion-free and thus $\tilde{\Gamma}$ is uniquely determined to be the Levi-Civita connection

In terms of the notation introduced in section 11.5, this amounts to the assertions

$$\begin{array}{ccccccc} T_{\lambda\mu\nu}(\tilde{\Gamma})=0 & \text{and} & \delta_{\tilde{\Gamma}}S[g,\tilde{\Gamma}]=0 & \Rightarrow & Q_{\lambda\mu\nu}(\tilde{\Gamma})=0 & \Rightarrow & \tilde{\Gamma}^{\lambda}_{\ \mu\nu}=\Gamma^{\lambda}_{\ \mu\nu} \\ Q_{\lambda\mu\nu}(\tilde{\Gamma})=0 & \text{and} & \delta_{\tilde{\Gamma}}S[g,\tilde{\Gamma}]=0 & \Rightarrow & T_{\lambda\mu\nu}(\tilde{\Gamma})=0 & \Rightarrow & \tilde{\Gamma}^{\lambda}_{\ \mu\nu}=\Gamma^{\lambda}_{\ \mu\nu} \end{array} \tag{20.113}$$

In either case, the metric equations of motion (20.112) then reduce to the vacuum Einstein equations.

In order to establish the assertions (20.113), we need two preparatory results. The first is that the generalisation of the formula (20.15),

$$\delta R_{\mu\nu}(\Gamma)=\nabla_\lambda\delta\Gamma^{\lambda}_{\ \mu\nu}-\nabla_\nu\delta\Gamma^{\lambda}_{\ \mu\lambda} \tag{20.114}$$

for the variation of the Ricci tensor in terms of the variation of the connection is

$$\delta R_{\mu\nu}(\tilde{\Gamma})=\tilde{\nabla}_\lambda\delta\tilde{\Gamma}^{\lambda}_{\ \mu\nu}-\tilde{\nabla}_\nu\delta\tilde{\Gamma}^{\lambda}_{\ \mu\lambda}+\left(\tilde{\Gamma}^{\rho}_{\ \nu\lambda}-\tilde{\Gamma}^{\rho}_{\ \lambda\nu}\right)\delta\tilde{\Gamma}^{\lambda}_{\ \mu\rho}\ . \tag{20.115}$$

The second is that when the connection is not the Levi-Civita connection, an expression like $\tilde{\nabla}_\lambda J^\lambda$ is not a total derivative in the integral, this being only true for the Levi-Civita connection thanks to the identity

$$\int\sqrt{g}d^4x\ \nabla_\lambda J^\lambda=\int d^4x\ \partial_\lambda(\sqrt{g}J^\lambda)\ . \tag{20.116}$$

Writing

$$\tilde{\Gamma}^{\lambda}_{\ \mu\nu}=\Gamma^{\lambda}_{\ \mu\nu}+C^{\lambda}_{\ \mu\nu}\ , \tag{20.117}$$

we have

$$\tilde{\nabla}_\lambda V^\mu=\nabla_\lambda V^\mu+C^{\mu}_{\ \nu\lambda}V^\nu \tag{20.118}$$

and, in particular,

$$\tilde{\nabla}_\lambda J^\lambda=\nabla_\lambda J^\lambda+C^{\lambda}_{\ \nu\lambda}J^\nu\ , \tag{20.119}$$

only the first term on the right-hand side giving rise to a total derivative.

What we are interested in is $g^{\mu\nu}\delta R_{\mu\nu}(\tilde{\Gamma})$, and with the above results and notation we can write this as

$$\begin{aligned} g^{\mu\nu}\delta R_{\mu\nu}(\tilde{\Gamma})=&g^{\mu\nu}\left(C^{\lambda}_{\ \rho\lambda}\delta C^{\rho}_{\ \mu\nu}-C^{\rho}_{\ \mu\lambda}\delta C^{\lambda}_{\ \rho\nu}-C^{\rho}_{\ \lambda\nu}\delta C^{\lambda}_{\ \mu\rho}+C^{\rho}_{\ \mu\nu}\delta C^{\lambda}_{\ \rho\lambda}\right) \\ &+\text{total derivative terms} \end{aligned} \tag{20.120}$$

Thus the variation of the action with respect to the $\tilde{\Gamma}$ is

$$\begin{aligned}\delta_{\tilde{\Gamma}} S[g,\tilde{\Gamma}] &= \int \sqrt{g} d^4x \; g^{\mu\nu} \left(C^{\lambda}_{\rho\lambda} \delta C^{\rho}_{\mu\nu} - C^{\rho}_{\mu\lambda} \delta C^{\lambda}_{\rho\nu} - C^{\rho}_{\lambda\nu} \delta C^{\lambda}_{\mu\rho} + C^{\rho}_{\mu\nu} \delta C^{\lambda}_{\rho\lambda} \right) \\ &= \int \sqrt{g} d^4x \; \left(g^{\alpha\beta} C^{\lambda\gamma}{}_{\lambda} + g^{\gamma\beta} C^{\alpha\lambda}{}_{\lambda} - C^{\beta\gamma\alpha} - C^{\alpha\beta\gamma} \right) \delta C_{\gamma\alpha\beta}\end{aligned} \tag{20.121}$$

If we were to consider arbitrary $\tilde{\Gamma}$ and hence unconstrained variations $\delta C_{\gamma\alpha\beta}$, the condition for the action to be stationary with respect to variations of $\tilde{\Gamma}$ would be

$$\delta_{\tilde{\Gamma}} S[g,\tilde{\Gamma}] = 0 \quad \Leftrightarrow \quad g^{\alpha\beta} C^{\lambda\gamma}{}_{\lambda} + g^{\gamma\beta} C^{\alpha\lambda}{}_{\lambda} - C^{\beta\gamma\alpha} - C^{\alpha\beta\gamma} = 0 \ . \tag{20.122}$$

However, these equations do not determine the $C^{\alpha}_{\beta\gamma}$ uniquely (we will explicitly parametrise this non-uniqueness below), and hence in this case the Einstein-Hilbert-like action (20.110) alone does not give rise to acceptable equations of motion for the fields.

The situation changes if one imposes some a priori constraints on the allowed $\tilde{\Gamma}$, and hence on their variations $\delta C_{\gamma\alpha\beta}$. We now consider separately the two cases mentioned above:

1. $\tilde{\Gamma}$ are restricted to be symmetric (torsion-free)

   In terms of the coefficients $C^{\gamma}_{\alpha\beta}$, this amounts to the condition

   $$C^{\gamma}_{\alpha\beta} = C^{\gamma}_{\beta\alpha} \tag{20.123}$$

   and the same condition should be imposed on their variations,

   $$\delta C^{\gamma}_{\alpha\beta} = \delta C^{\gamma}_{\beta\alpha} \tag{20.124}$$

   Thus, symmetrising appropriately, from (20.121) one obtains the constraints

   $$2 g^{\alpha\beta} C^{\lambda\gamma}{}_{\lambda} + g^{\gamma\beta} C^{\alpha\lambda}{}_{\lambda} + g^{\gamma\alpha} C^{\beta\lambda}{}_{\lambda} - 2C^{\beta\alpha\gamma} - 2C^{\alpha\beta\gamma} = 0 \ . \tag{20.125}$$

   Taking traces, once by contraction with $g_{\alpha\beta}$ and once by contraction with $g_{\alpha\gamma}$ (or, equivalently, with $g_{\beta\gamma}$), one obtains two linearly independent conditions on the traces $C^{\lambda\gamma}{}_{\lambda}$ and $C^{\alpha\lambda}{}_{\lambda}$ requiring both to vanish,

   $$\text{traces} \quad \Rightarrow \quad C^{\lambda\gamma}{}_{\lambda} = 0 \quad , \quad C^{\alpha\lambda}{}_{\lambda} = 0 \ . \tag{20.126}$$

   Then, upon symmetrisation, (20.125) reduces to the condition

   $$C^{\beta\alpha\gamma} + C^{\alpha\beta\gamma} = 0 \quad \Leftrightarrow \quad C_{\beta\alpha\gamma} + C_{\alpha\beta\gamma} = 0 \tag{20.127}$$

   As we have seen in (11.103) of section 11.5, this is precisely the condition that the non-metricity tensor is zero,

   $$Q_{\alpha\beta\gamma} = C_{\beta\alpha\gamma} + C_{\alpha\beta\gamma} = 0 \ , \tag{20.128}$$

i.e. that the connection is compatible with the metric.

Since we started off with a torsion-free (symmetric) connection, this means that the equations of motion fix the connection $\tilde{\Gamma}$ to be the Levi-Civita connection. Alternatively, (20.123) and (20.127) imply that $C_{\gamma\alpha\beta} = 0$ This concludes the proof of the first assertion in (20.113).

2. $\tilde{\Gamma}$ are restricted to be compatible with the metric

   This is largely analogous. In terms of the coefficients $C^{\gamma}{}_{\alpha\beta}$, metric compatibility amounts (as just recalled) to the condition

$$C_{\gamma\alpha\beta} = -C_{\alpha\gamma\beta} \tag{20.129}$$

   and the same condition should be imposed on their variations,

$$\delta C_{\gamma\alpha\beta} = -\delta C_{\alpha\gamma\beta} \tag{20.130}$$

   Thus

$$C_{\gamma\alpha\beta} = C_{\gamma(\alpha\beta)} + C_{\gamma[\alpha\beta]} = T_{(\alpha\beta)\gamma} + \tfrac{1}{2} T_{\gamma\alpha\beta} \tag{20.131}$$

   is the contorsion tensor. In this case, anti-symmetrisation of (20.121) leads to

$$g^{\alpha\beta} C^{\lambda\gamma}{}_{\lambda} - g^{\gamma\beta} C^{\lambda\alpha}{}_{\lambda} + C^{\beta\alpha\gamma} - C^{\beta\gamma\alpha} = 0 \ . \tag{20.132}$$

   Taking traces, one finds $C_{\lambda\gamma}{}^{\lambda} = 0$ (the other trace $C^{\lambda}_{\lambda\gamma}$ is identically zero because of anti-symmetry), and thus (20.132) reduces to

$$C^{\beta\alpha\gamma} - C^{\beta\gamma\alpha} = 0 \quad \Leftrightarrow \quad C_{\beta\alpha\gamma} = C_{\beta\gamma\alpha} \ , \tag{20.133}$$

   which is precisely the symmetry (no torsion) condition

$$T_{\beta\alpha\gamma} = C_{\beta\alpha\gamma} - C_{\beta\gamma\alpha} = 0 \ . \tag{20.134}$$

   Since we started off with metric-compatible connection, this means that the equations of motion fix the connection $\tilde{\Gamma}$ to be the Levi-Civita connection. Alternatively, (20.129) and (20.133) imply that $C_{\gamma\alpha\beta} = 0$. This concludes the proof of the second assertion in (20.113).

Alternatively, and perhaps somewhat more insightfully, one can first determine the general solution to the (under-determined) equation (20.122),

$$g^{\alpha\beta} C^{\lambda\gamma}{}_{\lambda} + g^{\gamma\beta} C^{\alpha\lambda}{}_{\lambda} - C^{\beta\gamma\alpha} - C^{\alpha\beta\gamma} = 0 \ , \tag{20.135}$$

and then analyse the properties of the solution and the consequences of imposing some conditions on $C_{\alpha\beta\gamma}$.[42] To disentangle this equation, we proceed as in the proof of the

[42] See e.g. A. Bernal et al., *On the (non-)uniqueness of the Levi-Civita solution in the Einstein-Hilbert-Palatini formalism*, `arXiv:1606.08756 [gr-qc]` and references therein.

uniqueness of the Levi-Civita connection in section 5.4 and take sums and differences of cyclic permutations of the above equation. Then one ends up with the equation

$$2C^{\alpha\beta\gamma} = g^{\alpha\beta}(A^\gamma + B^\gamma) + g^{\beta\gamma}(A^\alpha - B^\alpha) + g^{\gamma\alpha}(B^\beta - A^\beta) \ , \tag{20.136}$$

where

$$A^\alpha = C^{\alpha\beta}{}_\beta \quad , \quad B^\alpha = C^{\beta\alpha}{}_\beta \tag{20.137}$$

are two of the (a priori independent) traces of $C^{\alpha\beta\gamma}$. Performing either of these contractions in (20.136), one finds the condition

$$A^\alpha = B^\alpha \ , \tag{20.138}$$

and therefore

$$C^{\alpha\beta\gamma} = g^{\alpha\beta}A^\gamma \quad \Leftrightarrow \quad C^\alpha{}_{\beta\gamma} = \delta^\alpha_\beta A_\gamma \ . \tag{20.139}$$

Thus the general solution to the equation of motion (20.122) is

$$\tilde{\Gamma}^\alpha{}_{\beta\gamma} = \Gamma^\alpha{}_{\beta\gamma} + \delta^\alpha_\beta A_\gamma \ , \tag{20.140}$$

with A_γ an arbitrary covector.

This family of connections has the properties

$$\tilde{\nabla}_\gamma g_{\alpha\beta} = -2A_\gamma g_{\alpha\beta} \tag{20.141}$$

and

$$\tilde{\Gamma}^\alpha{}_{\beta\gamma} - \tilde{\Gamma}^\alpha{}_{\gamma\beta} = \delta^\alpha_\beta A_\gamma - \delta^\alpha_\gamma A_\beta \ . \tag{20.142}$$

It is now obvious that requiring either metric compatibility or the symmetry of the connection enforces $A_\gamma = 0$ and thus $\tilde{\Gamma} = \Gamma$.

Remarks:

1. When one couples either of these theories to matter, one will find the standard Einstein equations with source the usual matter energy-momentum tensor, provided that the minimally coupled matter action depends only on the metric and not on the connection. As we have seen, this is satisfied in the case of scalar or Maxwell gauge fields (for which the minimally coupled action in the usual setting could be written in such a way that it depends only on the metric but not on the derivatives of the metric). However, typically the connection appears explicitly in the action for spinors, and in this case variation of the matter action will produce a non-zero contribution to the torsion, say. Thus in that case the Einstein-Hilbert approach (no torsion as a kinematical constraint) and the Palatini approach (torsion determined dynamically) are inequivalent.

2. Within the present framework, it is not possible to relax both the no-torsion and the metricity constraints simultaneously and to simultaneously regain them dynamically, but one can attempt to achieve this either with the aid of additional auxiliary (Lagrange multiplier) fields, or "spontaneously" by adding potentials that force the connection in the ground state to either zero torsion or zero non-metricity.[43]

3. Once one contemplates and permits the presence of non-metricity and/or torsion, there are many more terms that one can in principle use to build an action (by using scalars built from the torsion and non-metricity tensors $T_{\mu\nu\lambda}$ and $Q_{\mu\nu\lambda}$ and their covariant derivatives). Thus, unless one imposes additional symmetry requirements, say, there is no good reason to focus attention exclusively on the Einstein-Hilbert-like action (20.110), and many generalisations of general relativity suggested and discussed in the literature can and should be either rejected or amended simply on these grounds.[44]

[43] See e.g. R. Percacci, *Geometry of Nonlinear Field Theories* for an exploration of some of these ideas.

[44] In his review of gauge theories of gravity, F. Hehl, *Gauge Theories of Gravity and Spacetime*, `arXiv:1204.3672 [gr-qc]`, also emphasises this: "Numerous pages of printed pages could be saved if our colleagues would [...] just motivate their choice of the unknown constants." In that review it is also pointed out that what is generally referred to (and I also called and will continue to call) the Palatini formalism should properly also be attributed to Einstein (1925).